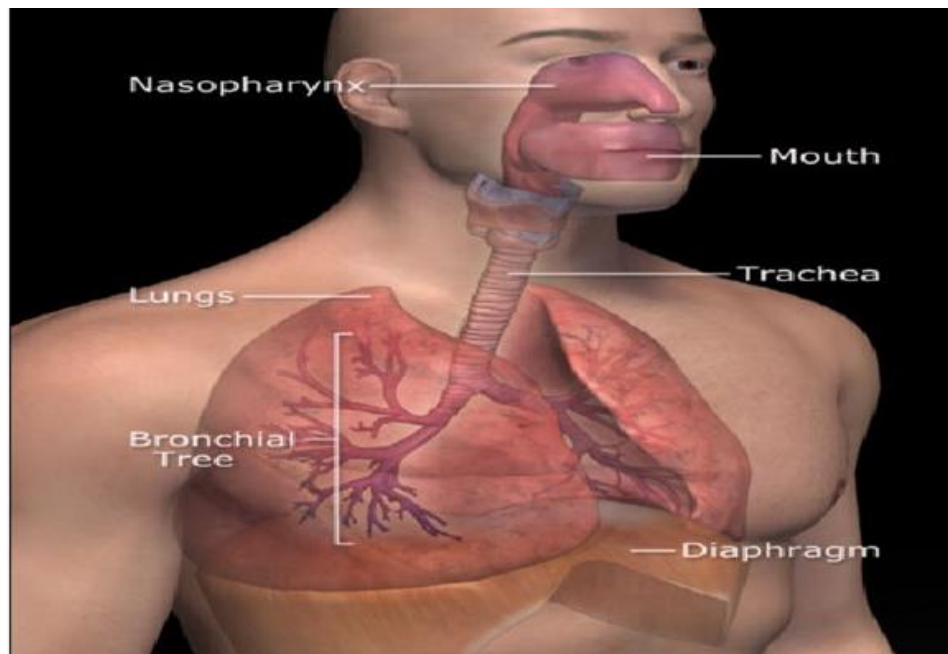
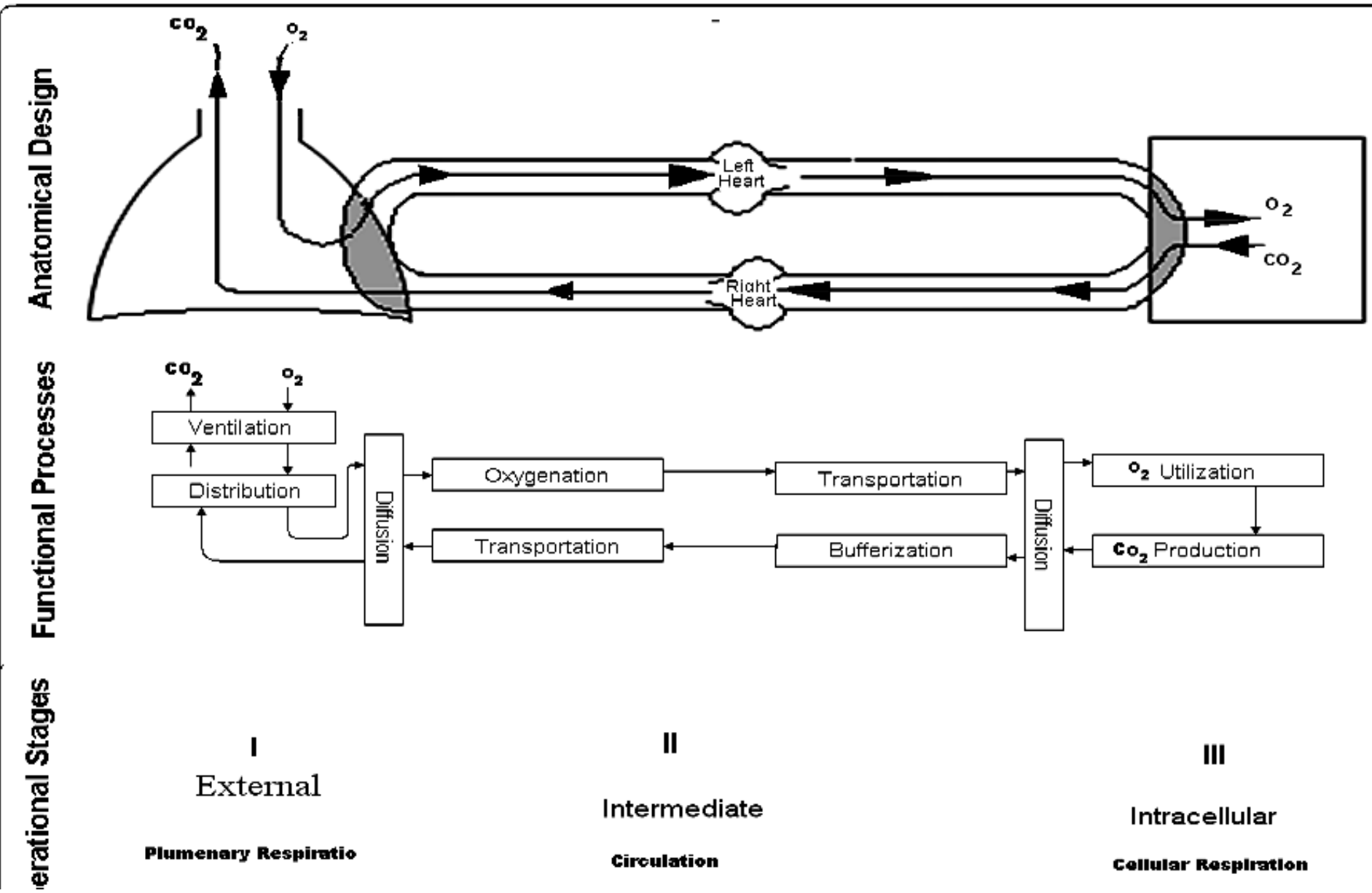


# Physiology of the Respiratory System



# Respiratory System

- Characteristics:
  - The prime function of the lung is to allow exchange of  $O_2$  for  $CO_2$
  - The lung also metabolizes some compounds + filters unwanted materials from the circulation
  - RS works in conjunction with CVS to facilitate exchange of blood gases
- Homeostatic roles:
  - Control of  $PaO_2$  +  $PaCO_2$
  - Control of hormone concentration
  - Removal of clot from the blood - filtration



# Respiratory System

## A. Primary Functions

- Extra-cellular respiration
  - Conduction of air to and from the terminal air spaces
- Intracellular respiration
  - Utilization of  $O_2$  and production of  $CO_2$

# Respiratory System

## B. Secondary Functions

### 1. Upper respiratory tract:

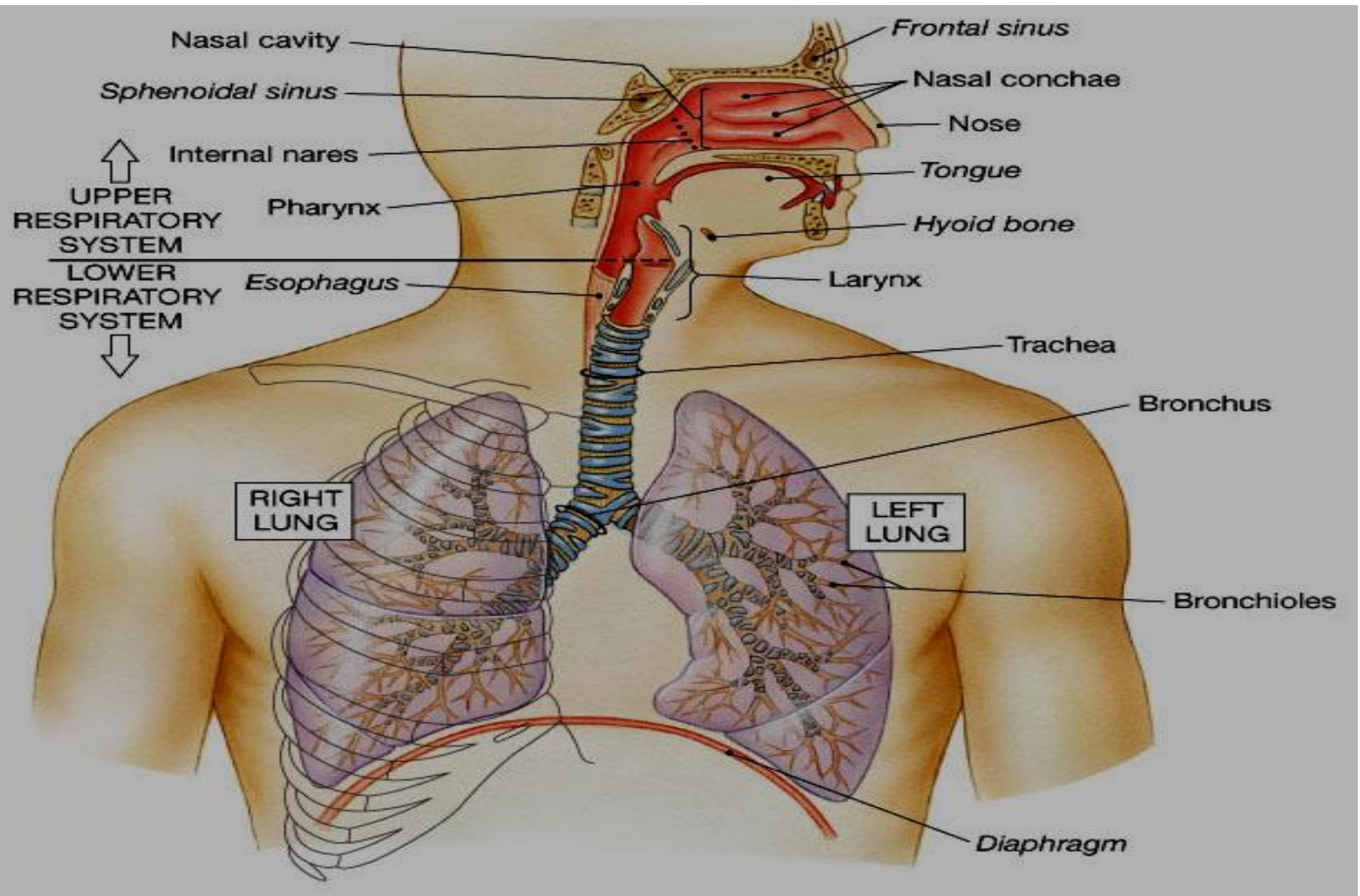
- Modification + conduction of inspired air
- Defense of the body against noxious stimuli like inhaled particles, acting as a filter to prevent blood clots
- Olfaction and phonation

### 2. Lower respiratory tract:

- Acid-base balance
- Formation of ACE through its vascular endothelium

## Functional Anatomy

- Components of the respiratory system:
  1. Upper air ways
  2. Lungs → a series of branching tubes consisting of
    - Conducting airways → trachea to terminal bronchioles
    - Respiratory airways → respiratory bronchioles to alveoli
    - Pleural sac
  3. Thoracic cage
  4. Muscles of breathing
  5. Control centers

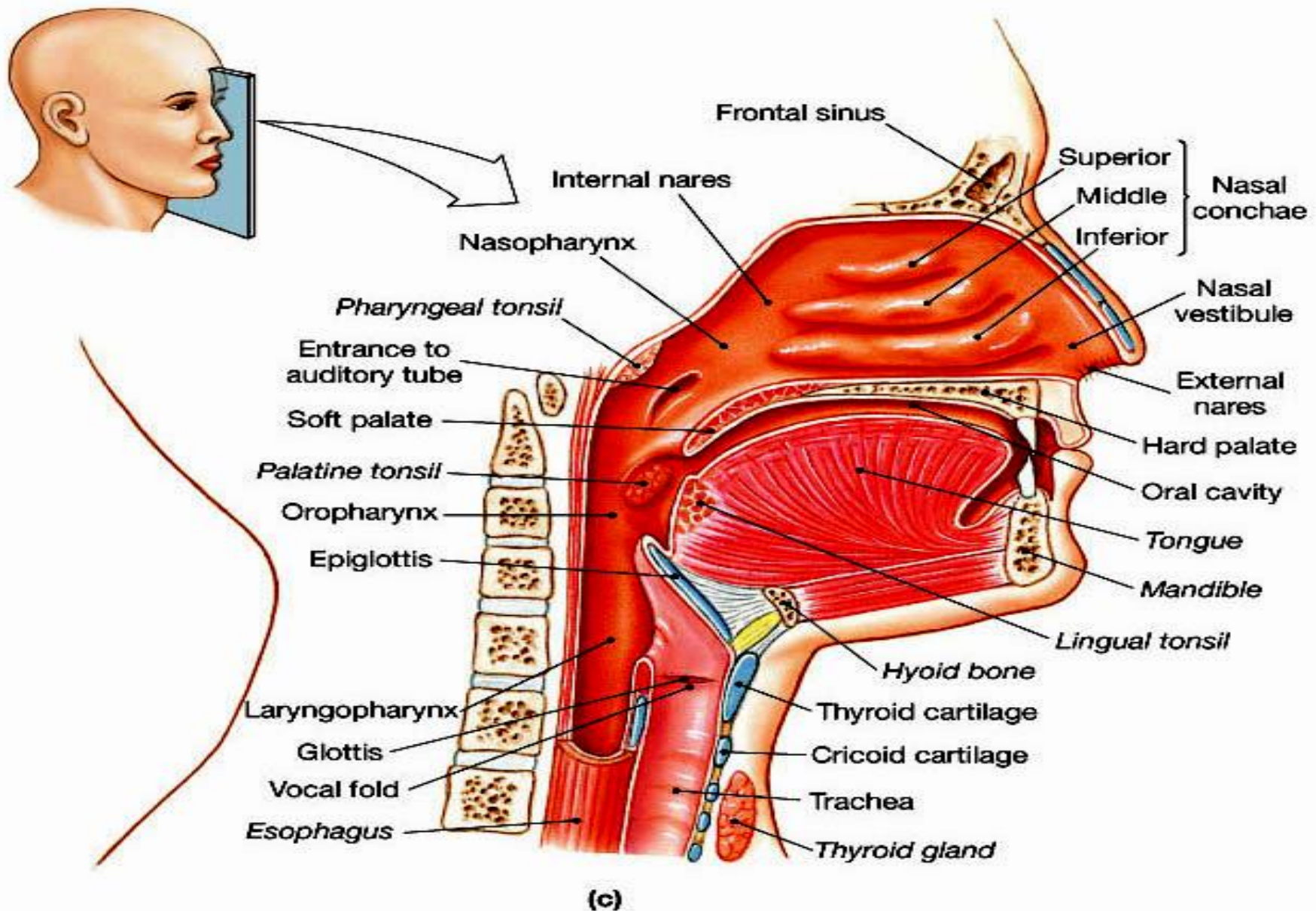


# Functional Anatomy

## 1. Conducting zone

- Nose (mouth as an alternative)
  - Nasal breathing is preferred route of breathing:
    - **Filtration** of particulate matter
    - Efficient **humidification** of air
- Pharynx
- Larynx
- Trachea
- Bronchi + Bronchioles + Terminal bronchioles

Nasal Cavity



# Nasal Cavity

- Components:

1. Nasal septum

- Partition between right and left halves of the nasal cavities

2. Nostrils (Nares)

- Two openings in the nasal cavities through which air enters

3. Vestibules

- The first ~1.5 cm of the conductive portion following the nostrils
- Lined with keratinized stratified squamous epithelium → for filtration

# Nasal Cavity

## 4. Nasal Conchae (Turbinates)

- Three bony projections on the lateral walls of the nasal cavities
- Increase the surface area of the nasal cavity
- Create turbulence in the stream of passing air
  - Facilitate air conditioning

5. Sebaceous glands → contribute to humidification of inspired air

6. Paranasal sinuses → drainage openings

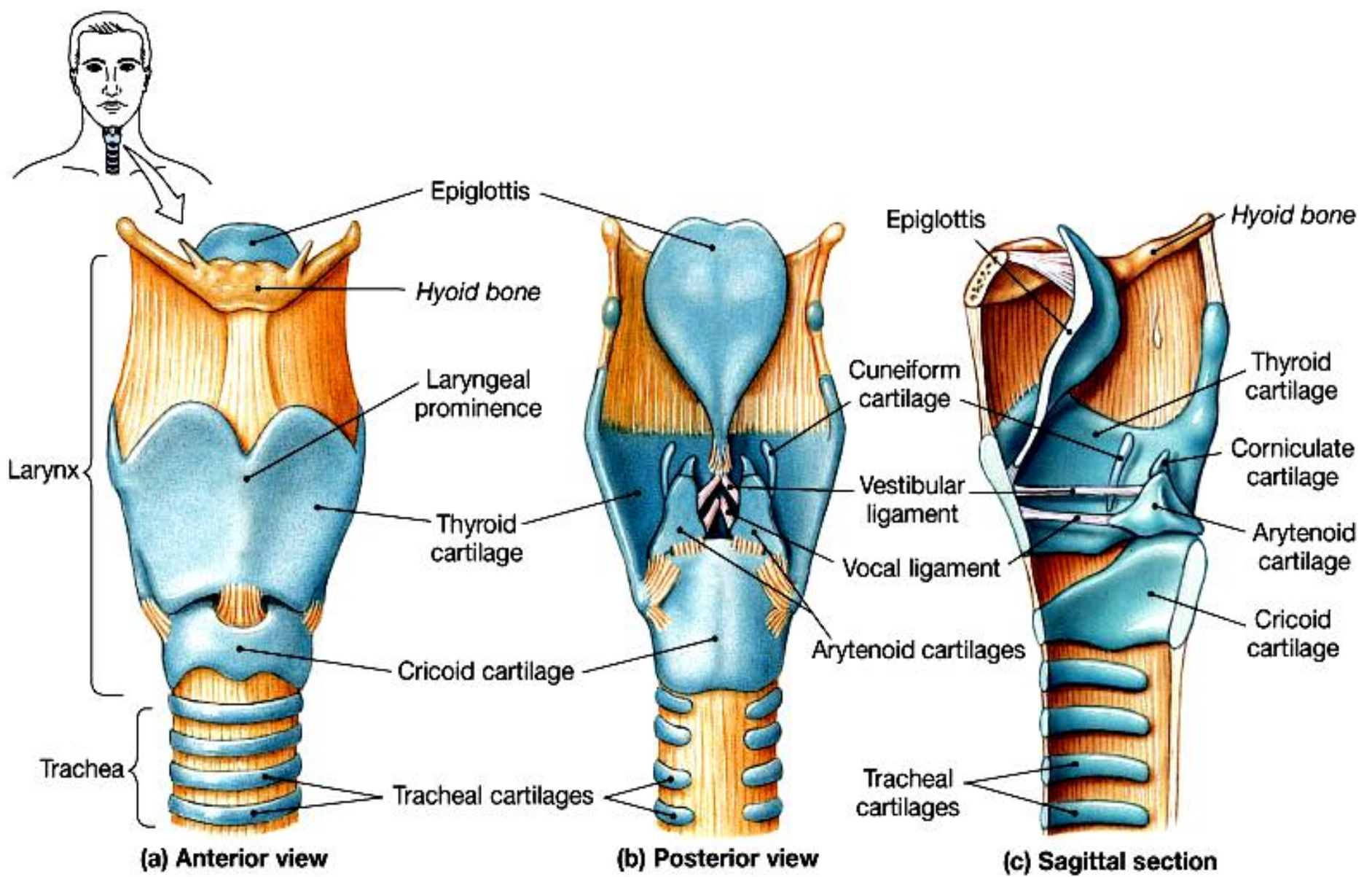
7. Olfactory receptors → receptor cells for the sense of smell

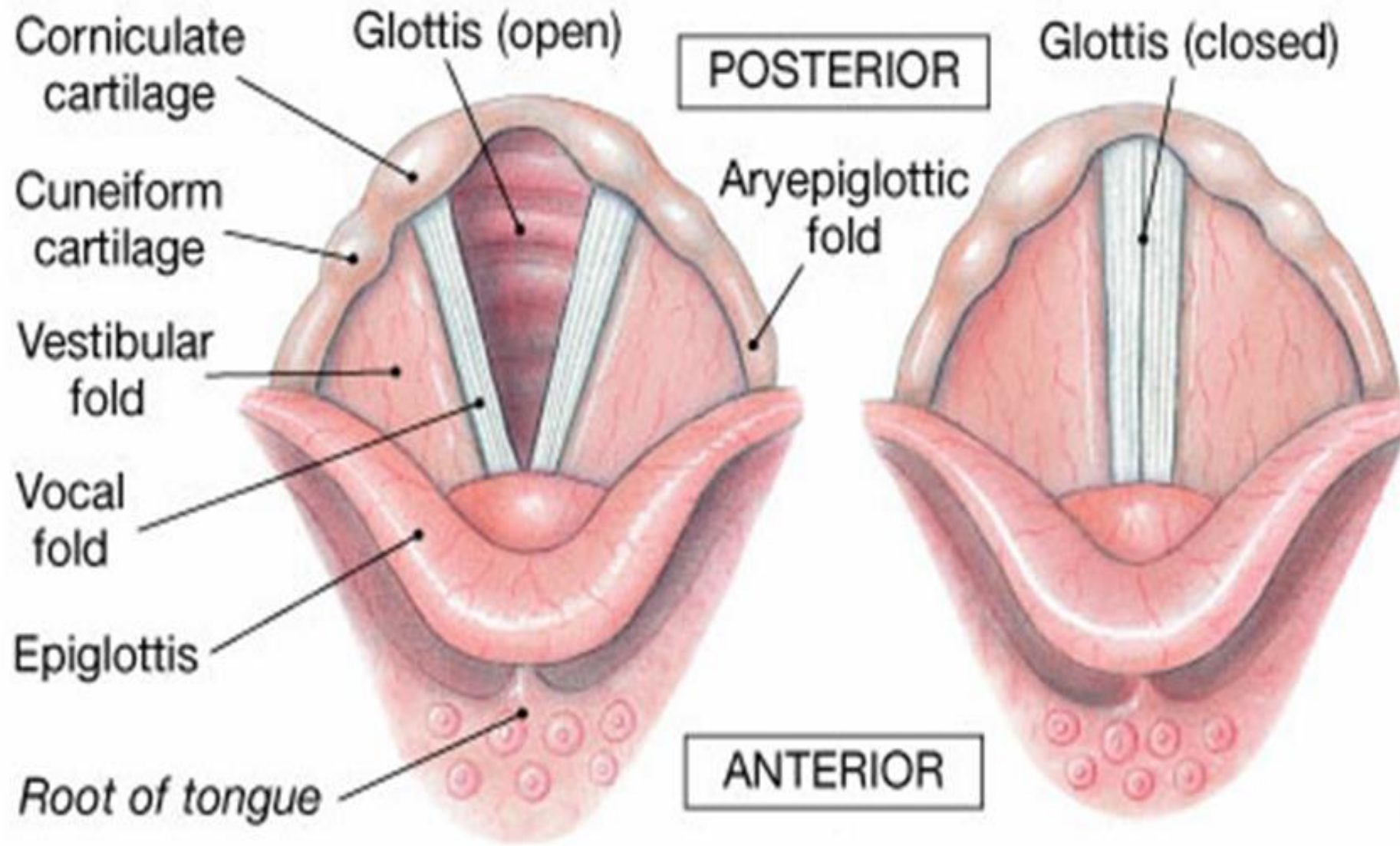
# Pharynx

- Cone-shaped passageway leading from the oral and nasal cavities to the esophagus and larynx
  - The pharynx chamber serves both as *respiratory and digestive pathway*
  - Thick muscle fibers and connective tissue attach the pharynx to the base of the skull and the surrounding structures
1. Oropharynx → Inlet to the laryngopharynx
  2. Laryngopharynx → Inlet to larynx
    - *Protective reflex during swallowing*
  3. Larynx → Connecting to nasal cavity and serving as first portion of air passage

# Larynx

- Voice box → Adam's Apple
- Connects pharynx to the trachea
- Consists of:
  - Thyroid cartilage
  - Epiglottis
  - Hyoid bone
  - Cricoid cartilage
  - Vocal cords
  - Glottis → aperture b/n vocal cords and surrounding area





**Larynx**

# Larynx

- Functions:
  - Connects the pharynx to trachea
  - Serves as the first portion of air passage
  - Its vocal folds provide **control of airflow** and **production of sound**
  - Acts as a **switching mechanism** to route air and food into the proper channels
  - Contracts rapidly to prevent food and liquids from entering the airway

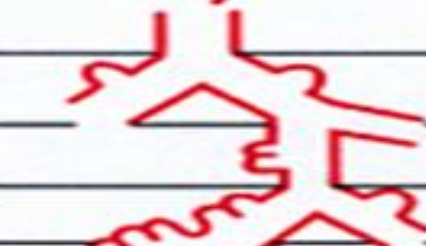
# Trachea

- A short tube ( $L \approx 10\text{-}12\text{cm}$ , diameter  $\approx 2\text{cm}$ )
- Lined by respiratory epithelium
- Extends from the larynx into the mediastinum
- Musculo-cartilaginous structure ( $10\text{cm} \times 2\text{cm}$ )
- Branches into two bronchi  $\rightarrow$  each breaks into 23 generations of progressively smaller bronchi, terminal bronchioles and respiratory bronchioles

**NB:** Trachea + Bronchi + Bronchioles  $\rightarrow$  the lower conducting airways

## Bronchi (Bronchial Tree)

- Conductive structures of down to ~1 mm
- Bronchi are extension of trachea (1<sup>st</sup>, 2<sup>nd</sup>, 3<sup>rd</sup> bronchi)
- Bronchi branch into Bronchioles (smaller generations up to terminal bronchioles)
- At the transition from bronchi to bronchioles, the epithelium changes to a ciliated columnar epithelium
- Respiratory bronchioles, alveolar ducts & alveolar sacs are the sites of gas exchange

|                                    |                         |                                                                                       |                                                                                      |    |
|------------------------------------|-------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----|
| Conducting zone                    | Trachea                 |                                                                                       | Z                                                                                    |    |
|                                    |                         |                                                                                       | 0                                                                                    |    |
|                                    | Bronchi                 |     | 1                                                                                    |    |
|                                    |                         |                                                                                       | 2                                                                                    |    |
|                                    |                         |                                                                                       | 3                                                                                    |    |
|                                    |                         |                                                                                       | 4                                                                                    |    |
|                                    | Bronchioles             |                                                                                       | 5                                                                                    |    |
|                                    |                         |                                                                                       | ↓                                                                                    |    |
| Terminal bronchioles               |                         |                                                                                       | 16                                                                                   |    |
| Transitional and respiratory zones | Respiratory bronchioles |                                                                                       |  | 17 |
|                                    |                         | 18                                                                                    |                                                                                      |    |
|                                    |                         | 19                                                                                    |                                                                                      |    |
|                                    | Alveolar ducts          |  | T <sub>3</sub>                                                                       | 20 |
|                                    |                         |                                                                                       | T <sub>2</sub>                                                                       | 21 |
|                                    |                         |                                                                                       | T <sub>1</sub>                                                                       | 22 |
|                                    | Alveolar sacs           |                                                                                       | T                                                                                    | 23 |

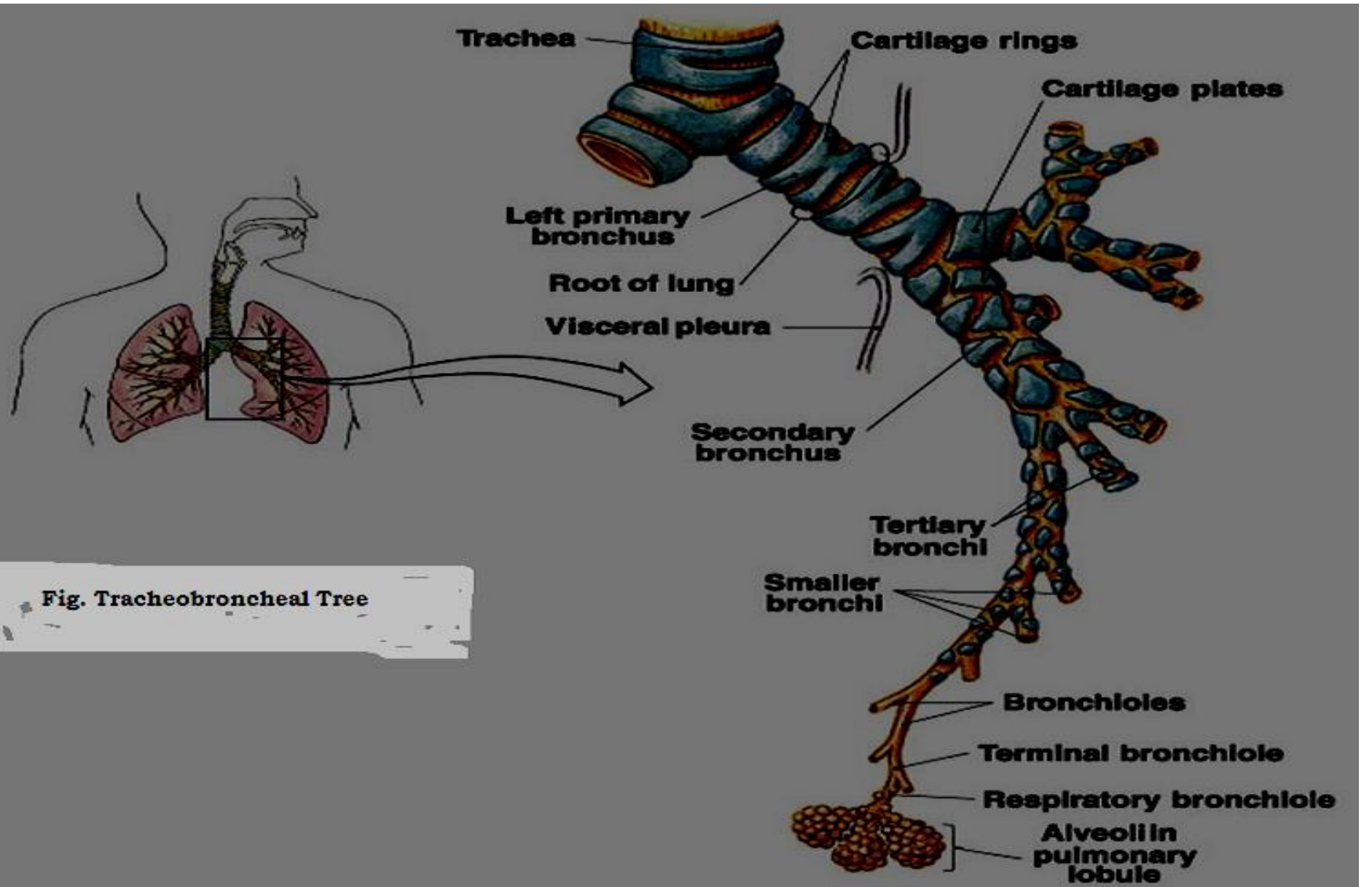


Fig. Tracheobroncheal Tree

# Cilia

- Hair-like projections lining the respiratory passages to the end of the bronchioles
- The mucus in cilia forms an “escalator” (mucus escalator) →  
*Mucociliary blanket of the air ways*
- When one breathes, essentially inhales a lot of junk containing particulate matter and micro-organisms sticking to mucus
- Particulate matter sticking to mucus is moved by cilia to pharynx → thus keeps the lungs clean

# Cilia

- The air passages and the lungs are defended against bacterial infection by this mucus escalator
- Ciliary activity can be reduced by **smoking + hypoxia** leading to lung infection
  - a single cigarette smoked → non-motility of cilia for several hours
    - Non-motility of cilia + secretion of mucus → accumulation of secretion → partial or complete **airway obstruction**

# The Exchanging Zone

- Consists:
  - Respiratory bronchioles
  - Alveoli + Alveolar duct + Alveolar sac
- Exchange of  $O_2$  for  $CO_2$  occurs through **alveolo-capillary (respiratory) membrane**
- Alveolo-capillary membrane is made of several layers and consists of fluid film lining the alveoli and capillary endothelial cells
- Alveoli ( $\approx 300 \times 10^6$ ) make an exchanging area of about 40 x the surface area of the body ( $40 \times 1.75 m^2 \approx 70 m^2$ )

# The Lungs

- Lower conducting airways
  - Gas exchange areas
  - Pulmonary interstitium and
  - Its blood vessels
- 
- form the lungs
- Alveoli are small out-pouching of the walls of the respiratory bronchioles that serve as sites of gas exchange in the lungs

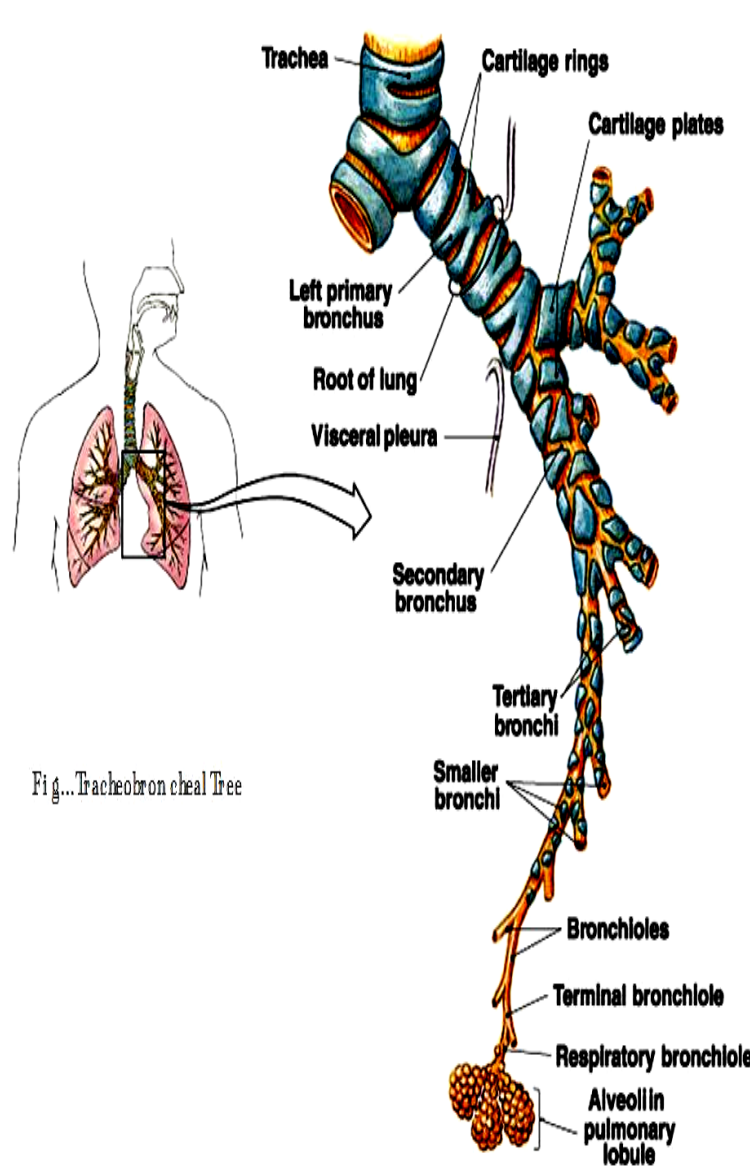
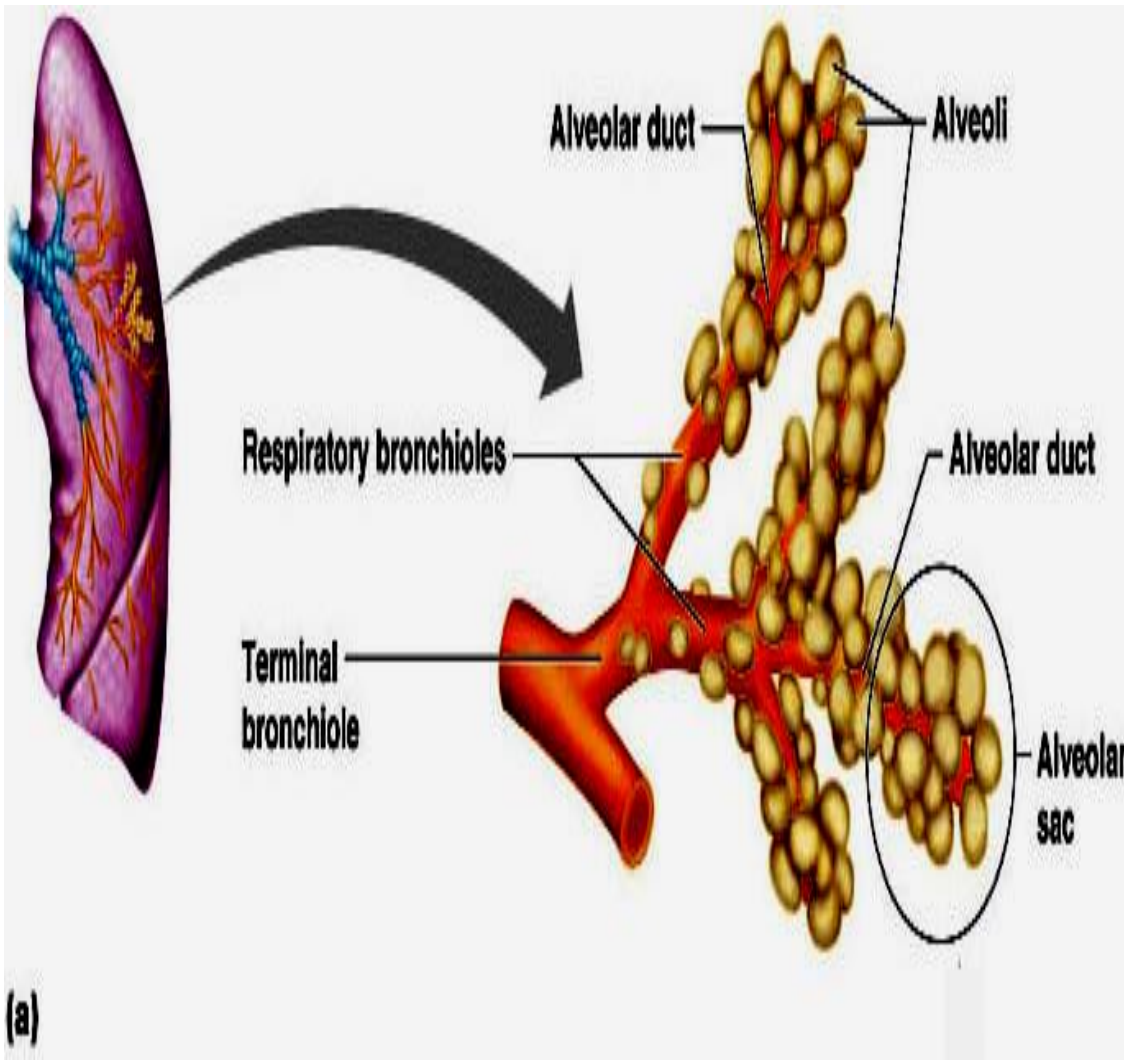
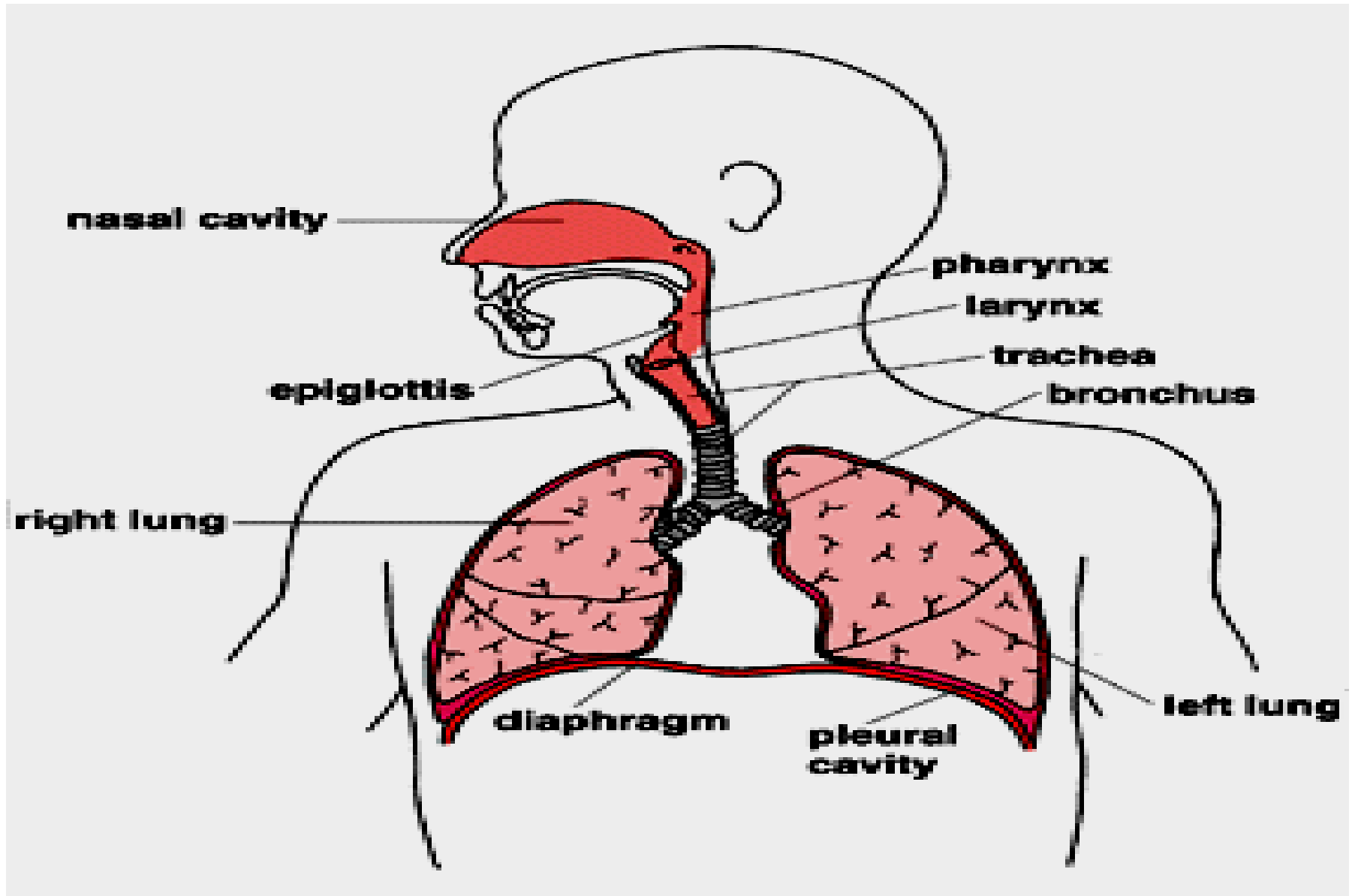


Fig...Tracheobronchial Tree



# The Lungs

- **Right lung:**
  - Separated into three lobes → superior + middle + inferior
  - Accounts for  $\approx$  55% of the pulmonary function
- **Left lung** → separated into upper and lower lobes
  - Accounts for  $\approx$  45% of the pulmonary function
- Each lung is surrounded by two layers of **serous membrane (pleurae)**
  - Visceral and parietal layers are separated by a small cavity filled with **pleural fluid** → assists in producing smooth breathing movements by acting as a **lubricant**



# Alveoli

- Consists of:
- **Type I Pneumocytes**
  - Major lining cells with simple thin squamous epithelium
  - Gas exchange takes place
- **Type II Pneumocytes**
  - Less in number and constitute thicker granulocytes
  - Responsible for the production of surfactants
- **Type III Pneumocytes**
  - Large phagocytic macrophage cells found in alveolar cavities
  - Keep alveolar surfaces sterile by removing debris and microbes

# Alveoli

- Alveolar Pneumocytes are:
  - Supplied with blood vessels + lymphatics + nerves
  - Connected by open pores
    - Allow air pressure throughout the lung to be equalized  
*(alveolar interdependence)*
- Form part of the respiratory (alveolo-capillary) membrane

## Pulmonary Blood Flow

- Pulmonary circulation:
  - Receives the entire cardiac output via **pulmonary artery**
  - Pulmonary artery branches in concert with airways
  - Blood flows in sheet-like rather than in discrete capillaries to maximize gas exchange
  - Pulmonary blood pressure is **25/8mmHg** (*mean* = 10mmHg)
  - The driving pressure from heart to heart is 10mmHg (10% of the systemic circulation)
  - Hence right heart and pulmonary vessels are thin walled

# Pulmonary Blood Flow

- **Bronchial arteries**

- Receive  $\approx 1\%$  of the cardiac output (50 ml/min)
- Nourishes the trachea down to the terminal bronchioles + pleura and hilar lymph nodes (more abundant)
- Have **high vascular resistance** → controlled by **bronchomotor nerve**

- **Pulmonary veins**

- Unlike arteries they do not follow the air ways
- Situated within interlobular loose connective tissue septa
- Receive blood from many lung units
- They are thin walled elastic structures

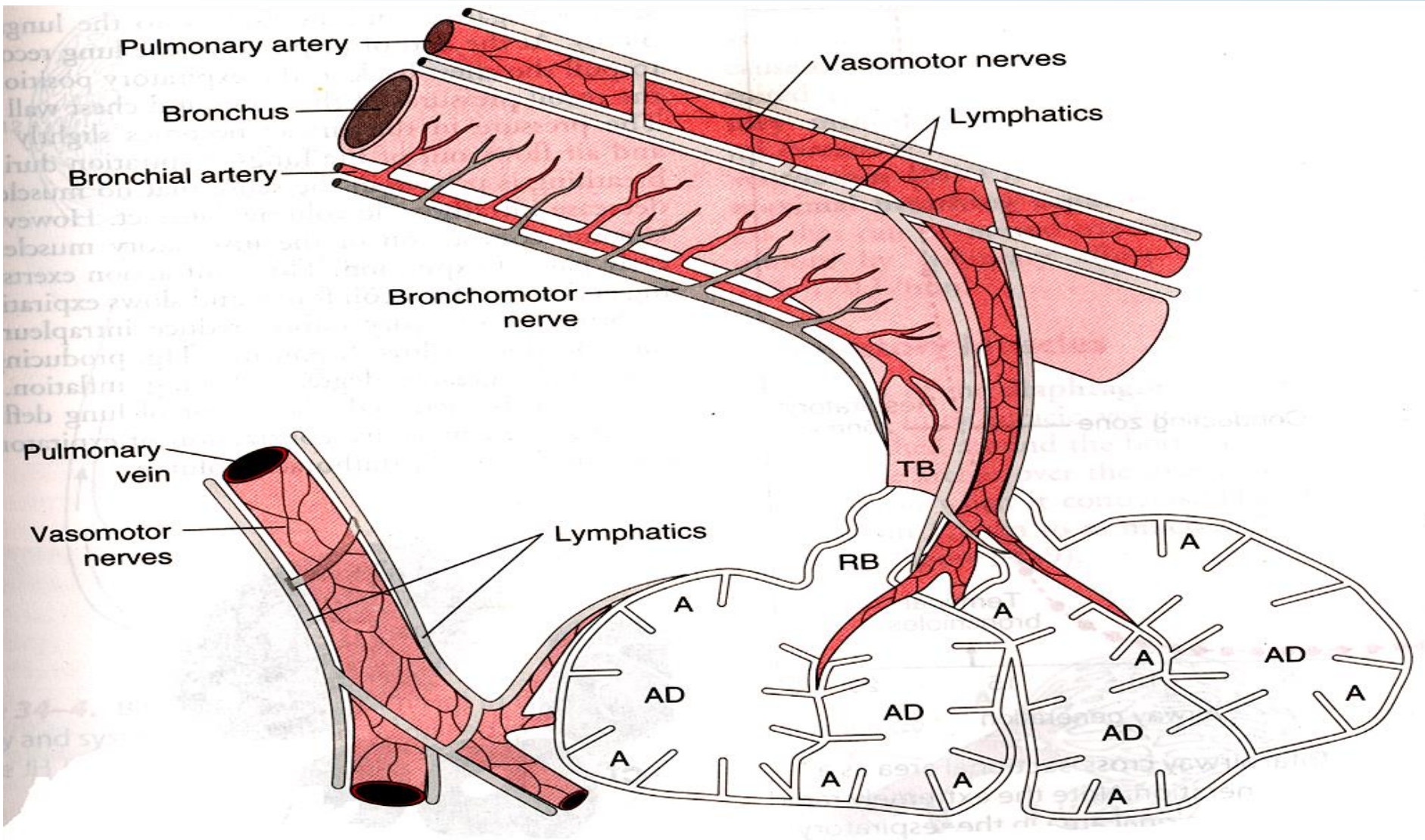
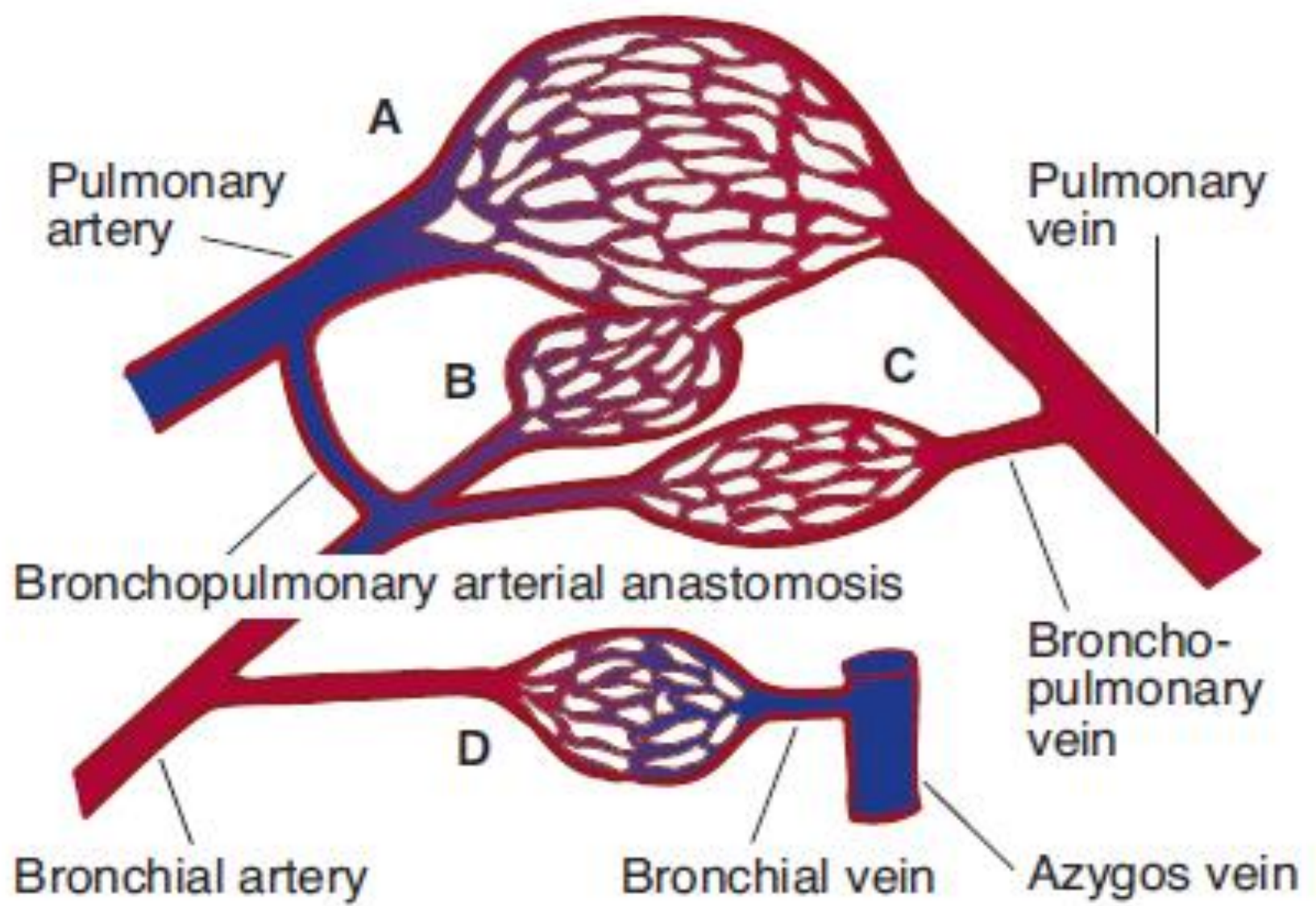


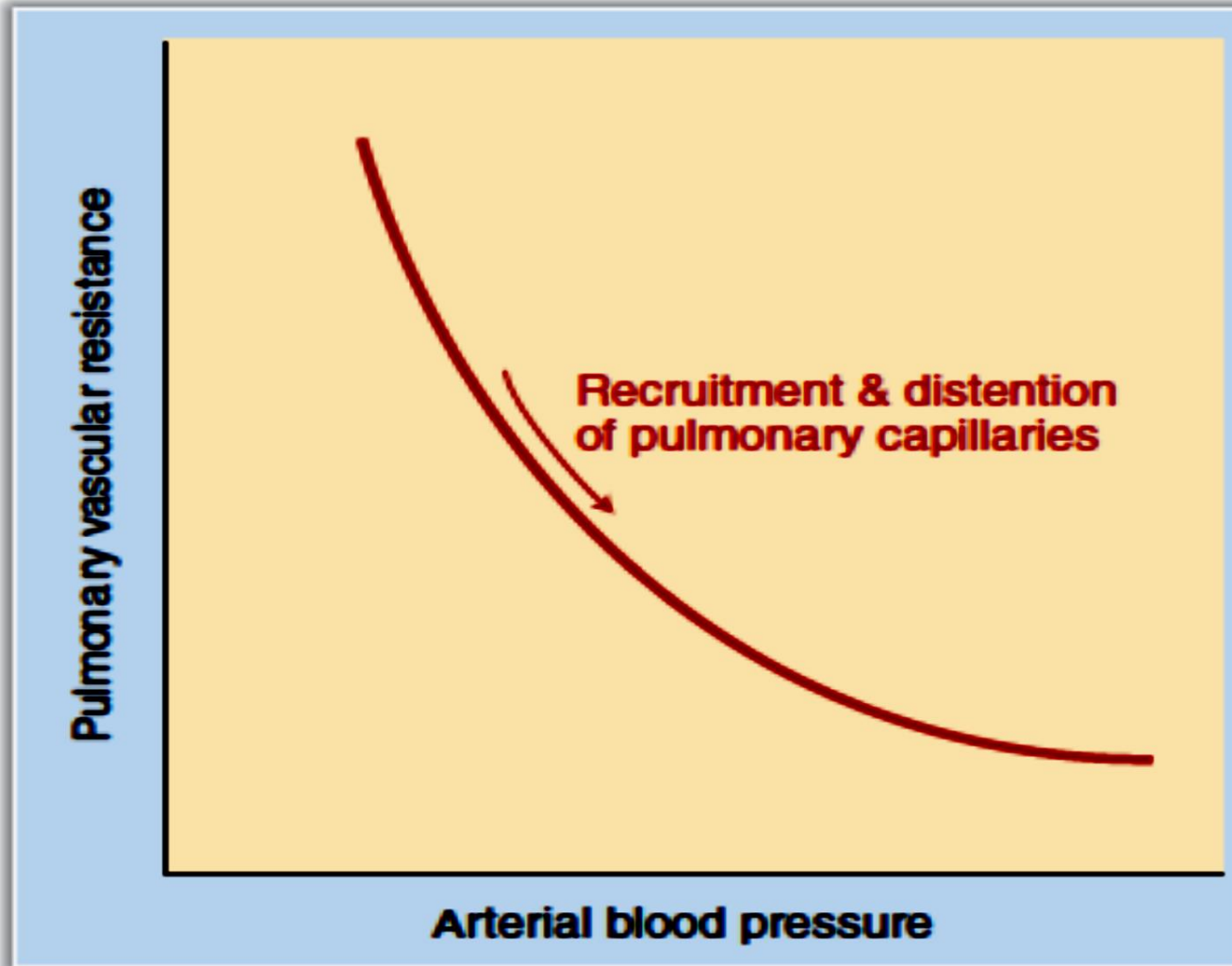
Fig.3a. Structure of the Lung: A, Alveolus; AD, Alveolar Duct; Respiratory bronchiole; TB, Terminal Bronchiole.



# Pulmonary Blood Flow

- **Pulmonary capillaries**

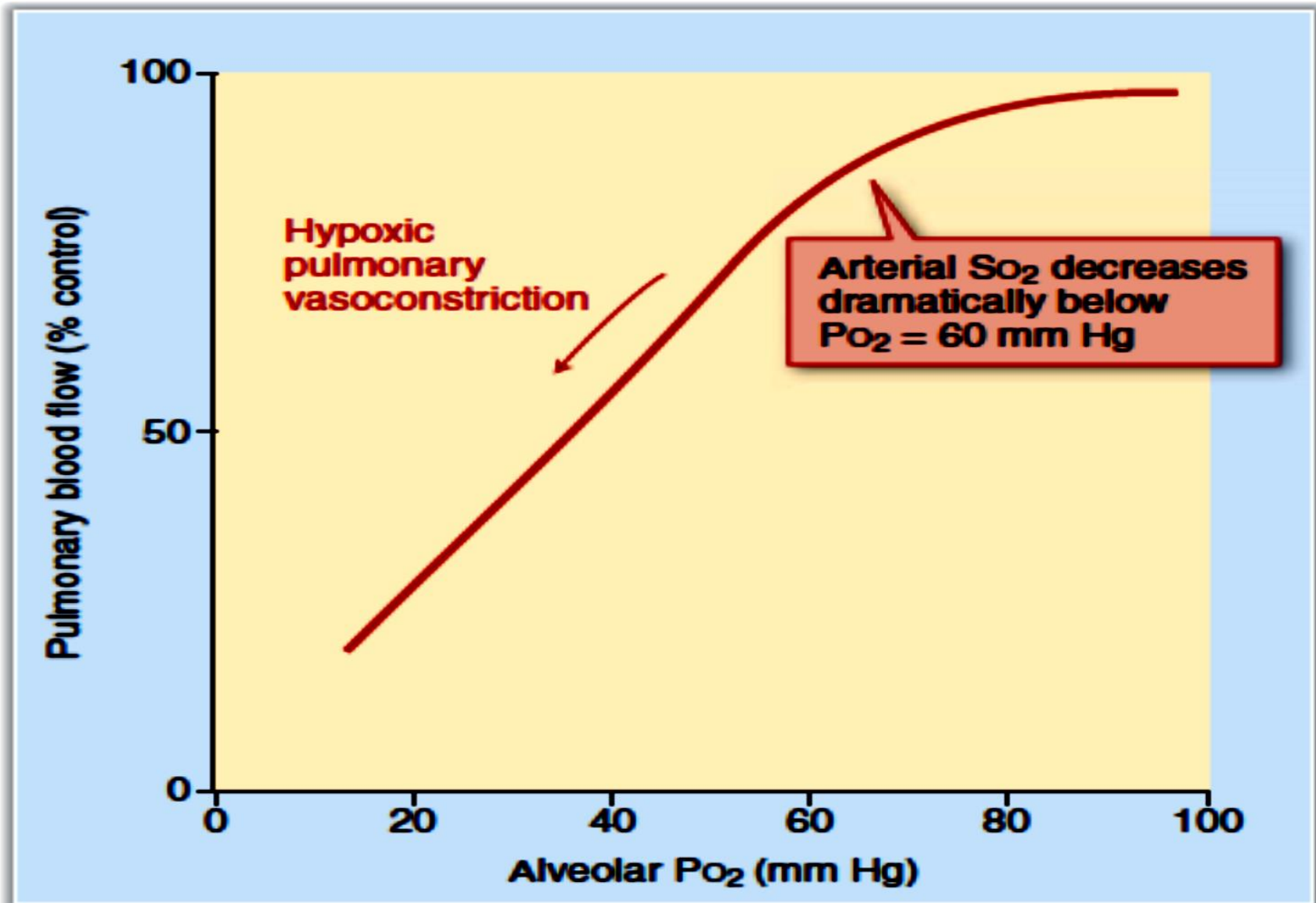
- Resting capillary blood volume = Stroke volume = 70ml
- Maximum capillary blood volume (heavy exercise)  $\approx$  4-7 fold
- This increment is accommodated by:
  - **Recruitment**  $\rightarrow$  increasing the number of open capillaries
  - **Distending** all the capillaries
- Recruitment and distention decrease pulmonary vascular resistance  $\rightarrow$  little rise in pulmonary arterial pressure

**A.**

**Effects of pulmonary blood pressure (cardiac output) on pulmonary vascular resistance**

## Pulmonary Blood Flow

- **Hypoxic pulmonary vasoconstriction**
  - Low alveolar  $PO_2$  ( $PAO_2$ ) causes pulmonary vasoconstriction and reduces pulmonary blood flow
  - The effect is marked at  $PAO_2$  lower than 60mmHg, where arterial oxygen saturation ( $SO_2$ ) begins to sharply decrease on the oxyhemoglobin dissociation curve
  - Rapid ascent to high altitude also causes pulmonary vasoconstriction (hypertension)



## Pulmonary Blood Flow

- **Hypoxic pulmonary vasoconstriction: Significance**
  - In **fetus**, it shunts blood away from the lungs
  - **After birth**, the first breath (alveolar oxygenation) dilates pulmonary vessels and allow oxygenation
  - **After birth**, hypoxic vasoconstriction shunts blood away from poorly ventilated alveoli → **improving V/Q (ventilation/perfusion)**

## Pulmonary Blood Flow

- **Regional blood flow:**
  - Alveolar capillaries are distended by the capillary blood pressure, but compressed by alveolar air pressure
  - Hence, if alveolar air pressure  $>$  capillary blood pressure  $\rightarrow$  capillaries close and there is no blood flow
  - This results in 3 possible zones of pulmonary blood flow

# Pulmonary Blood Flow

- **Zone-1**

- Alveolar air pressure  $>$  arterial pressure
  - This causes compression of pulmonary capillaries (at the apex)
- *No blood flow* during all portions of the cardiac cycle
- This is the *alveolar dead space* (ventilated but not perfused)
- Low ABP (hemorrhage) and high alveolar ventilation (ventilator) worsens zone 1

# Pulmonary Blood Flow

- **Zone-2**

- Systolic arterial pressure  $>$  alveolar air pressure  $>$  diastolic arterial pressure
- Vessels are partially constricted and flow is more limited
  - *Intermittent blood flow*

- **Zone-3**

- Alveolar capillary pressure  $>$  alveolar air pressure at all times
- Area of lungs with largest blood flow (all lower areas of the lung)
- *Continuous blood flow*

# Pulmonary Blood Flow

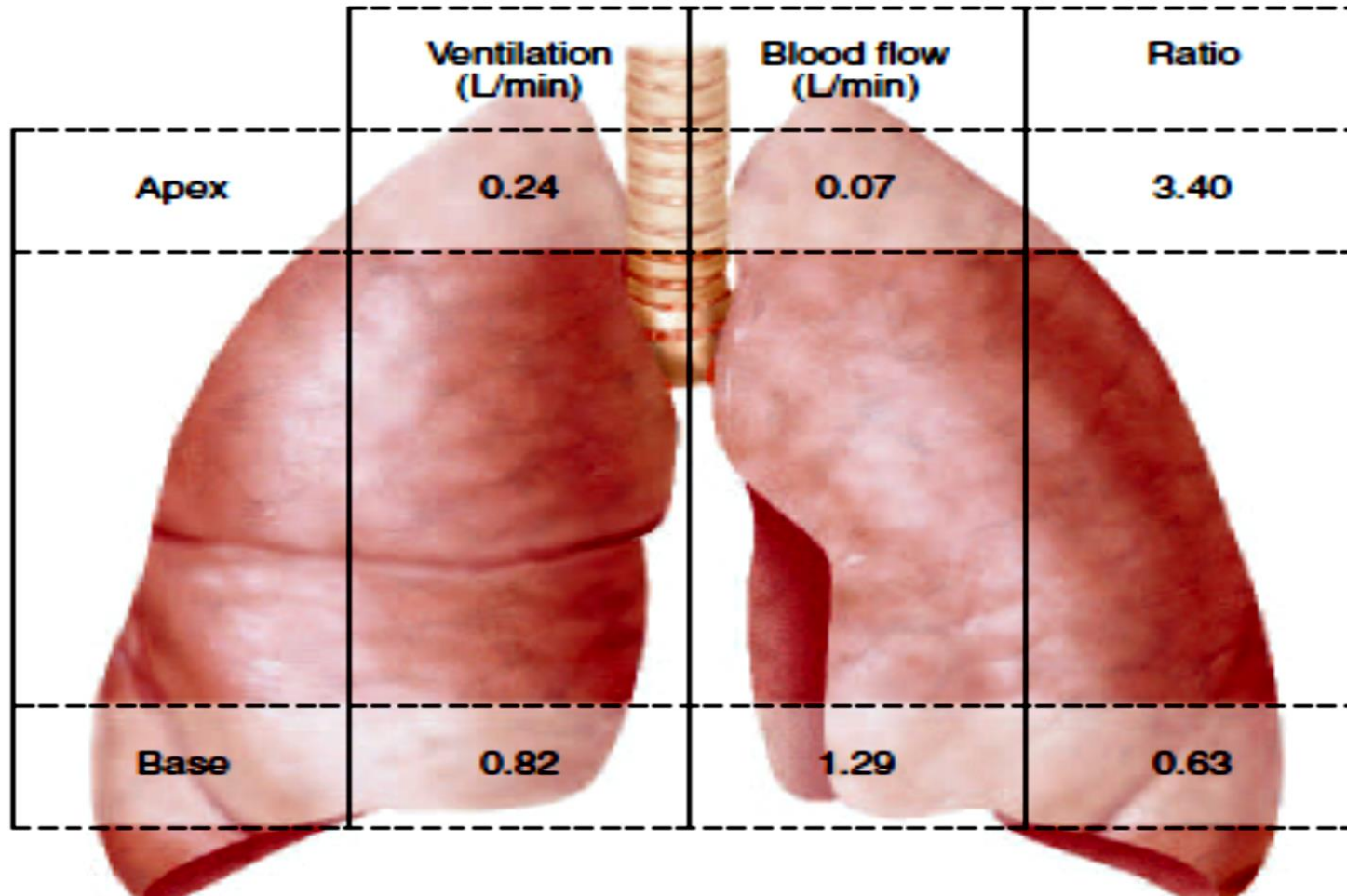
- **Regional blood flow:**

- Normally, in the upright position,  $VA/Q$  varies b/n
  - 3-5 in the apex and
  - 0.3-0.5 in the bottom of the lungs

- **Lying position:**

- $\approx$  all part of the lung is a few cm above the level of the heart
- Hence, alveolar capillary pressure is always higher  $\rightarrow$  blood flow is continuous throughout the lung
- Most efficient gas exchange occurs when  $VA/Q = 1.0$

## Ventilation/Perfusion



high  $VA/Q$  is caused by either excessive ventilation or inadequate blood flow

---

# **The Mechanics of Breathing**

---

# The Mechanics of Breathing

- Involves all structures that provide the forces of respiratory movement in the thoracic cage
- Force for the movement of lungs and chest wall is generated by **respiratory muscles** to overcome the **works of breathing**
- Respiratory muscle stimulation → muscle force (F) → Pressure ( $P=F/\text{area}$ ) → External power (ventilation)
- The forces of breathing are applied to the **thoracic cage**

# The Mechanics of Breathing

- **Thoracic cage:**
  - Makes a solid casing around the lung to protect it and prevent its collapse by the elastic recoil of its tissues
  - Contains **external and internal intercostal muscles** filling the gap between the ribs

# The Mechanics of Breathing

- **Innervation:**
  - **Chemoreceptors** from aortic and carotid bodies in the neck region and the brain stem → Respiratory centers (medulla and pons)
  - **Diaphragm** → by Phrenic nerves arising from C3-5
  - **Intercostal muscles** → by Intercostal nerves (thoracic segments)
  - **Tracheo-broncheal tree** → by autonomic nerves:
    - PNS ( $M_3$ ) → bronchial constriction and secretions
    - SNS:  $\beta_2$  → bronchial dilatation
      - $\alpha_1$  → decrease bronchial secretions

# The Mechanics of Breathing

- Work of Breathing:

1. *Compliance/elastic work*

- Required to expand the lungs against the lung and chest elastic forces

2. *Tissue resistance work*

- Required to overcome the viscosity of the lung and chest wall structures

3. *Airway resistance work*

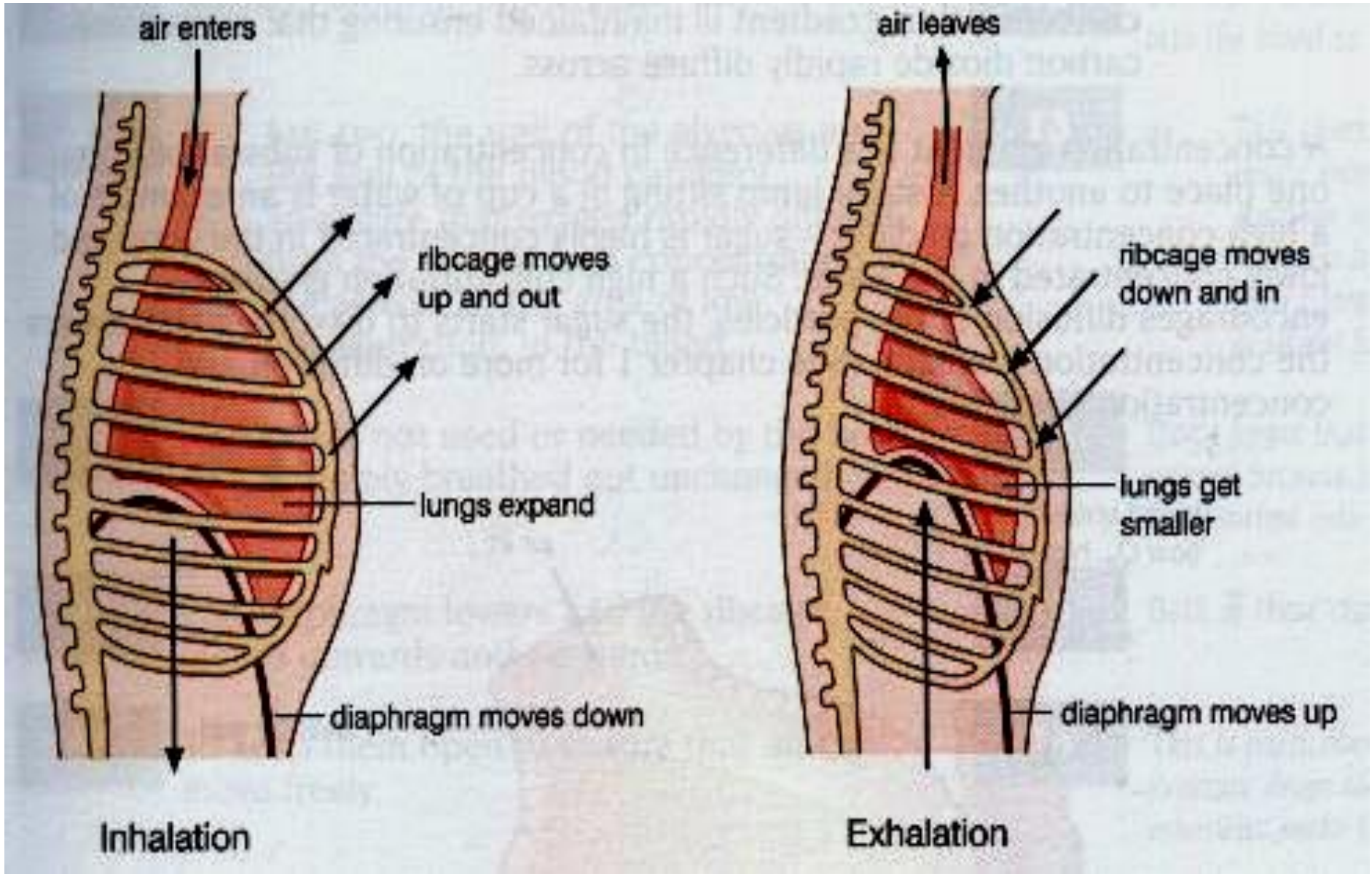
- Required to overcome airway resistance to the movement of air into the lungs

# The Mechanics of Breathing

| Groups of muscles                                                                                                                                           | Resistances to be overcome during inspiration                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Diaphragm                                                                                                                                                | <b>Elastic resistance (65%)</b> <ul style="list-style-type: none"> <li>Elastic recoil of lungs</li> <li>Elastic recoil of thorax</li> </ul>                                  |
| 2. Intercostals                                                                                                                                             |                                                                                                                                                                              |
| 3. Auxiliary muscles: <ul style="list-style-type: none"> <li>Sternocleidomastoid</li> <li>Scalenes</li> <li>Pectoralis</li> <li>Latissimus dorsi</li> </ul> | <b>Non-elastic resistance</b> <ul style="list-style-type: none"> <li>Airway resistance (<b>20%</b>)</li> <li>Tissue resistance of lungs &amp; thorax (<b>15%</b>)</li> </ul> |

# Inspiration

- Applying expansion force to the chest cage through inspiratory muscles:
  - Contraction of diaphragm → ↑ **vertical dimension** of the chest cage
  - Contraction of external intercostal muscles → ↑ **antero-posterior axis** of the chest cage
- Active movement of the chest cage → Passive movement of lungs through linkages of thin film of fluid between the lungs and chest wall → chest cage enlarges → ↑ volume of thorax → more negative pleural pressure (P<sub>pl</sub>) → **Lung expansion**



# Inspiration

## Active



**Nerve impulse**



**Contraction of:**

- Diaphragm
- External intercostals
- Accessory muscles

## **Contraction of Diaphragm**



↑ Vertical dimension of thorax by its descent (1.5cm up to 7cm)



**Contributes 70% of TV**

## **Contraction of External intercostals**



↑ Antero-posterior dimension of thorax



**Contributes 30% of TV**

## **N.B**

- **Accessory muscles** are involved when **TV > 800ml/breath** during inspiration

# Expiration

- Reduction of chest cavity → ↓volume of thorax → ↓-ve Ppl → ↓lung vol.

| Passive at rest                                                                |                                                                           | Active during exercise                    |                                                                                                |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------|
| Elastic recoil of                                                              | Surface tension effect of:                                                | Nerve impulses → Contraction of:          |                                                                                                |
| Stretched tissues in thorax, lungs, abdominal muscles<br>↓<br><b>30% of TV</b> | Fluid lining alveoli and respiratory bronchioles<br>↓<br><b>70% of TV</b> | Expiratory muscles (internal intecostals) | Anterior abdominal muscles including pelvic floor muscles<br>↓<br>Upward movement of diaphragm |

# Lung Compliance

- The ease with which the lungs distend
- Stretchability of the lung tissue and chest wall
- The extent to which the lungs expand for each unit change in transpulmonary pressure
  - Relates  $\uparrow$  volume (strain) to  $\uparrow$  pressure (stress)
  - $\Delta$  in distending pressure across the lung-alveolus and the pleural space
- Varies with the degree of lung inflation
- Normal compliance of both lungs  $\approx 200 \text{ ml/cmH}_2\text{O}$
- Compliance of lung + thorax  $\approx 110 \text{ ml/cmH}_2\text{O}$

# Lung Compliance

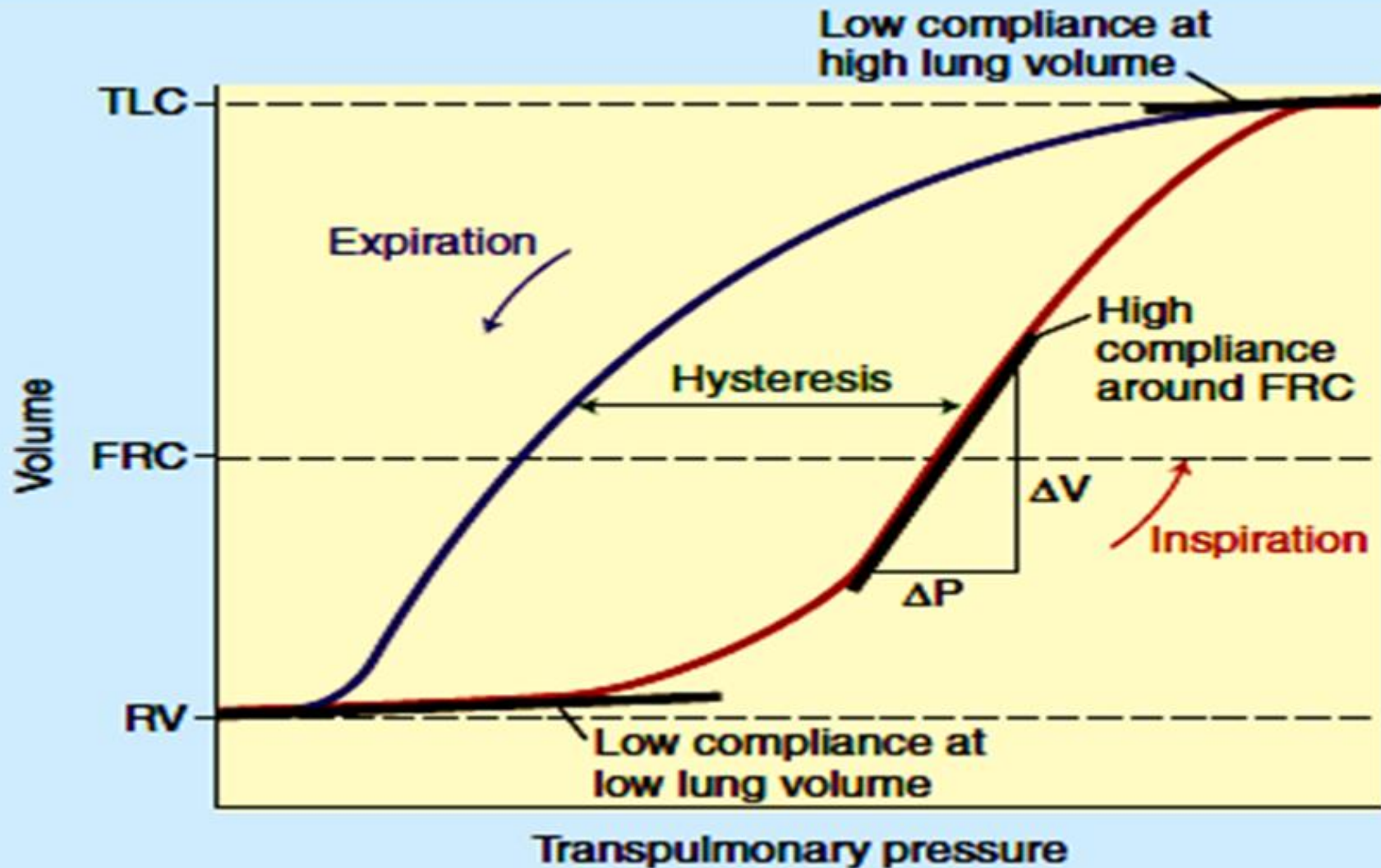
- **Hysteresis:**
  - P-V relationship showing the difference between expiratory & inspiratory compliance
  - The difference (Hysteresis) is due to surfactants which repel each other and resist the tendency of alveoli to be smaller
    - This maintain alveoli open during expiration → **more time for gas to diffuse**

# Lung Compliance

- **Inspiratory compliance curve**
  - Surfactants are dispersed, thus surface tension  $\uparrow$
  - Recoil pressure is high
  - Collapse pressure is high
- **Expiratory compliance curve**
  - Surfactant molecules are condensed which decreases surface tension

# Lung Compliance

- Hysteresis:
  - Deep Inspiration → expansion of alveoli → ↑surface area of fluid → dispersion of surfactant → ↓surfactant effect → ↑surface tension → decreased compliance → prevent alveolar overdistension and rupture
  - Expiration → recoil of alveoli to smaller size → ↓ fluid surface area → condensed surfactant → ↑ surfactant activity → ↓surface tension → increased compliance → prevent lung collapse



**FRC reflects the equilibrium between the lung and chest wall recoil forces**

# Lung Compliance

- At FRC, Compliance is maximum
  - FRC is associated with smallest work of breathing
  - At FRC, airway pressure is zero and all respiratory muscles are relaxed
- Lungs tend to collapse at all lung volumes from RV to TLC:
  - At **high lung volumes**, Compliance decreases due to **positive recoil** pressure in the airways after lung is inflated (lung tend to collapse)
  - At **small lung volumes**, Compliance decreases due to **collapsing of the lung acini**

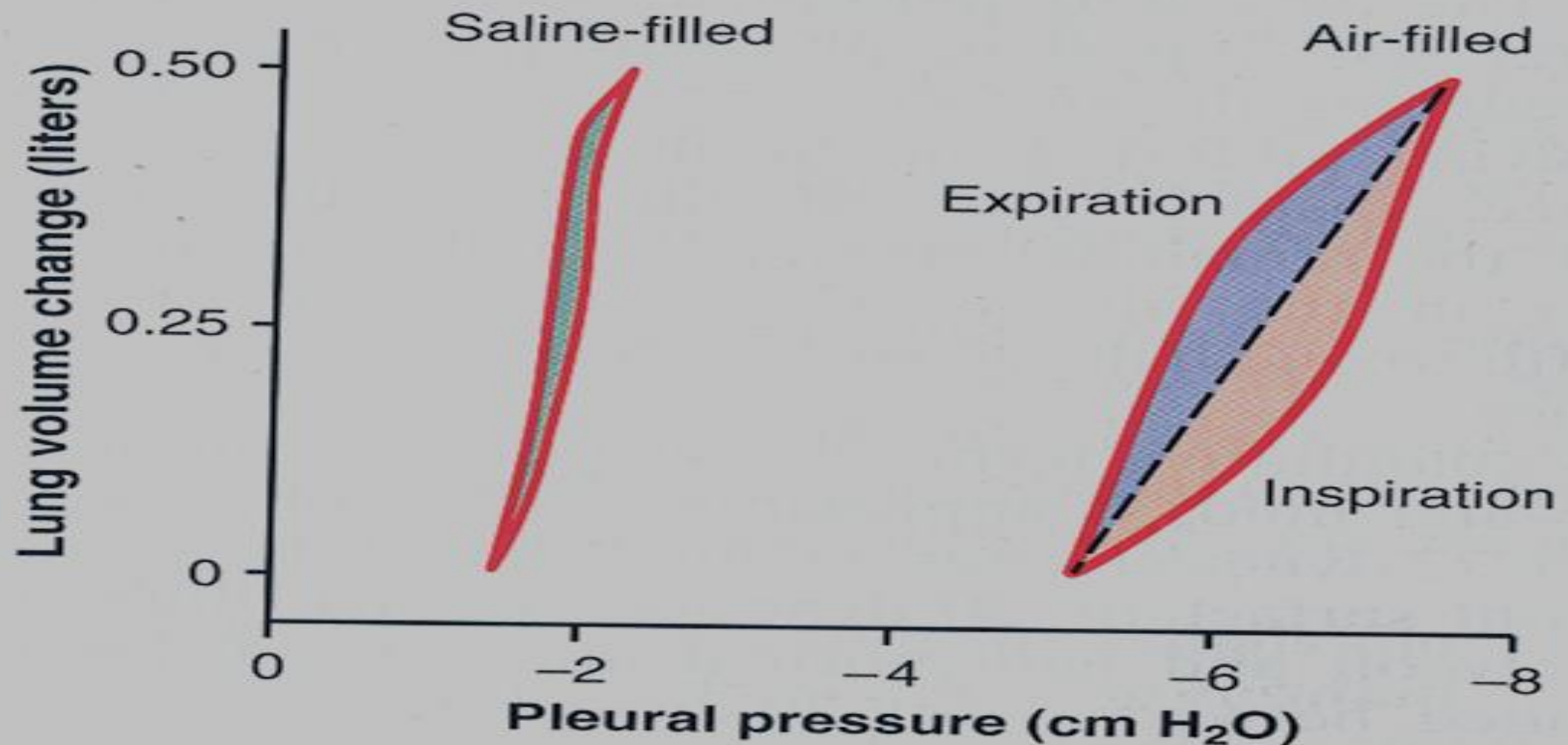
# Lung Compliance

- Two forces determine the compliance characteristics:
  1. Elastic and collagen fibers of the lung itself
    - Contracted and kinked at small volume
    - Stretched and unkinked at large volume
  2. Surface tensions of fluid inside the alveoli
    - Lung filled with air:
      - Creates an interface of alveolar water with the alveolar air
      - This increase the surface tension
    - Lung filled with saline takes away the air-water interface

# Lung Compliance

- Surface tension of water molecules
  - Water molecule in interface with air have strong attraction for one another
  - Hence there is a tight contractile membrane of water molecules around the surface in a droplet
- Surface tension in Alveoli
  - Surface tension is created by water in fluid surrounding the alveoli
  - Adjacent water molecules on surface of alveoli cause attractive forces that are much more stronger than those b/n liquid and gas

## Compliance (saline filled vs. air filled lung)



# Surfactants

- **Characteristics:**
  - Surface-active lipoprotein that lines the alveoli
  - Reduces surface tension and stabilizes the alveoli
- **Source** → Type-II Pneumocytes
- **Composition:**
  - Dipalmytoilphosphatidylcholine (DPPC): main
  - Surfactant apoproteins
  - Calcium ion

# Surfactants

- **Actions:**
  - Greatly reduce surface tension from 70 dynes/cm to 10 dynes/cm
  - Increase dispensability of the lungs
  - Reduce recoil force of the lungs
  - Prevent alveolar collapse by balancing pressure between small and large alveoli (*Alveolar interdependence*)
  - Reduce the work of breathing by reducing the collapse pressure created in the alveoli
  - Reduce formation of pulmonary edema

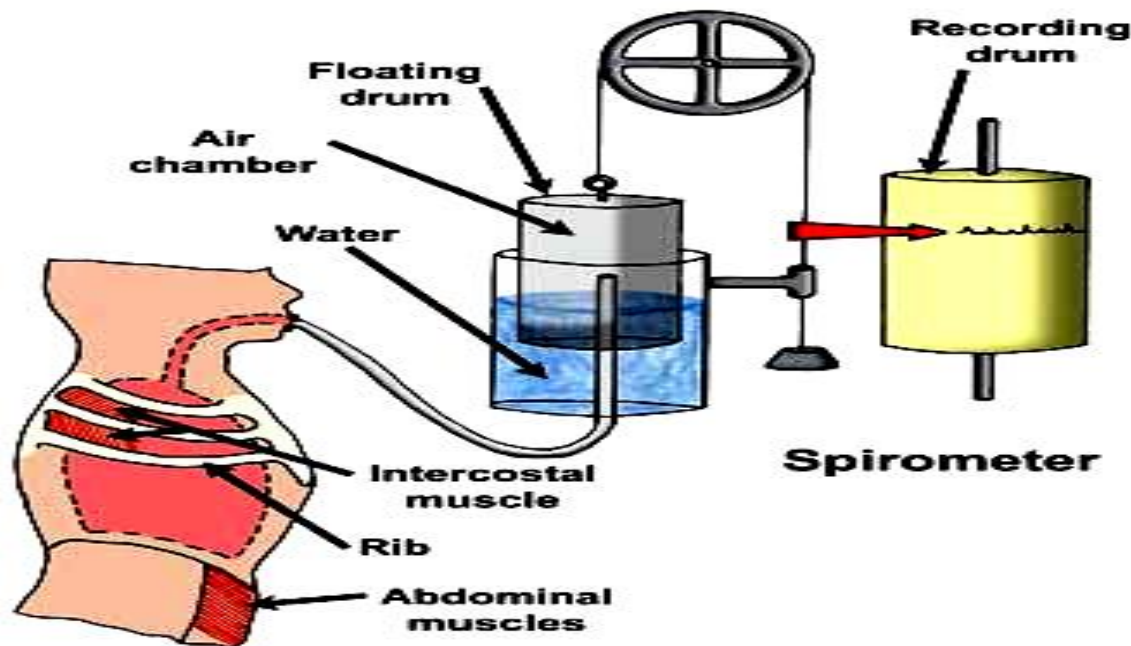
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# **Ventilation**

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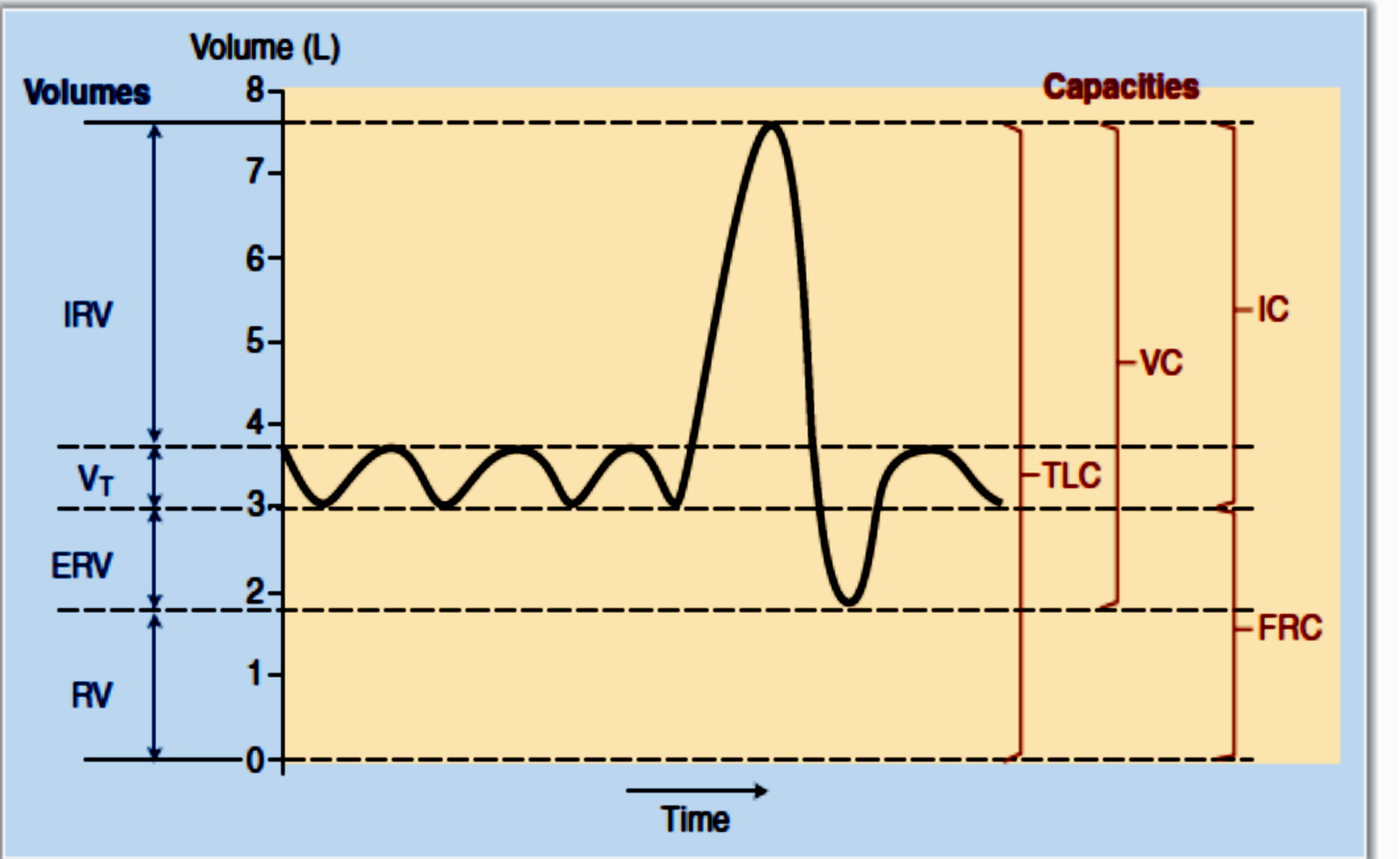
# Ventilation

- Gas exchange between the atmosphere and the lungs
- Mechanical events producing gas movement in the lungs resulting in  $\text{CO}_2$  elimination and  $\text{O}_2$  uptake
- Measured by a **Spirometer**

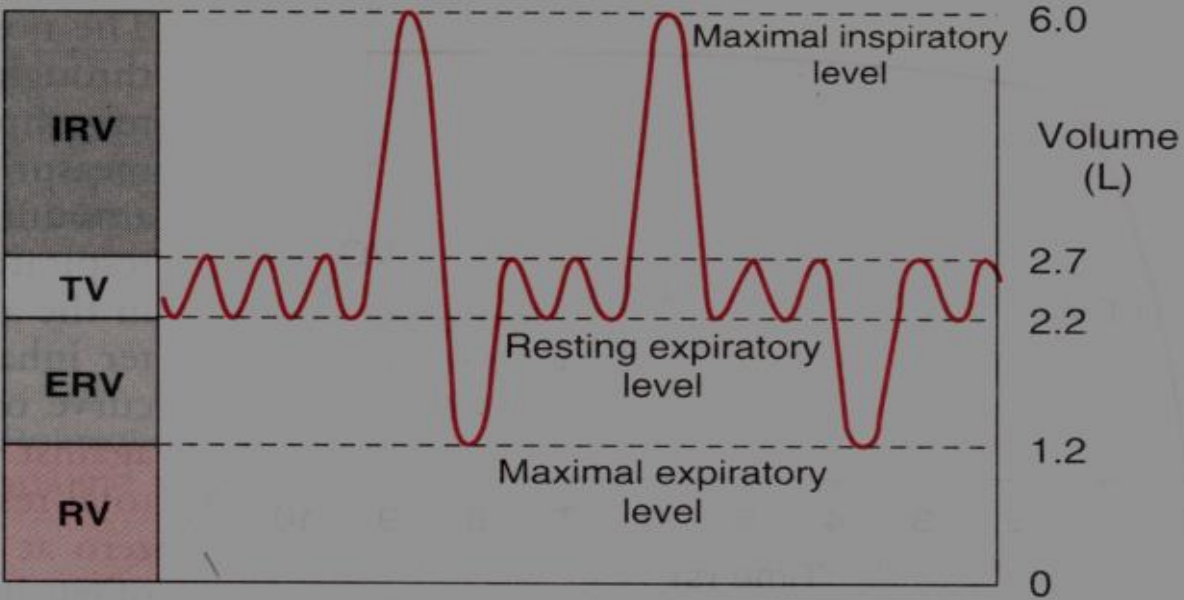
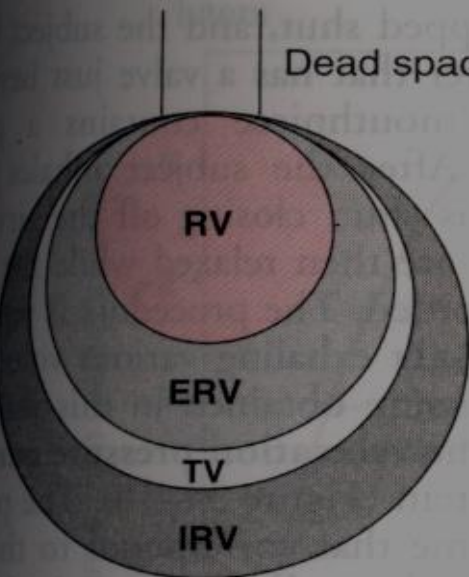


| Primary Lung Volume                 | Definition                                                                                                                                 |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Tidal volume ( $V_T$ )              | Volume of a quiet breath.                                                                                                                  |
| Inspiratory reserve volume (IRV)    | Volume that can be inspired above $V_T$ .                                                                                                  |
| Expiratory reserve volume (ERV)     | Volume that can be expired from the end of a tidal expiration.                                                                             |
| Residual volume (RV)*               | Volume remaining in the lung after maximal expiration.                                                                                     |
| Primary Lung Capacity               |                                                                                                                                            |
| Total lung capacity (TLC)*          | (IRV + $V_T$ + ERV + RV): Total volume of gas in the lung after maximal inspiration.                                                       |
| Vital capacity (VC)                 | (IRV + $V_T$ + ERV): Volume of maximal expiration from a maximal inspiration.                                                              |
| Inspiratory capacity (IC)           | (IRV + $V_T$ ): Maximal inspiration from the end of tidal expiration.                                                                      |
| Functional residual capacity (FRC)* | (ERV + RV): Volume of gas in the lungs at the end of tidal expiration<br>(reflects equilibrium between lung and chest wall recoil forces). |

\*These capacities cannot be measured directly by spirometry because they include residual volume.



| <b>Lung Volume /Capacity</b> |  <b>(male)</b> |  <b>(FEMALE)</b> |
|------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>TV</b>                    | <b>700 ml</b>                                                                                   | <b>500 ml</b>                                                                                       |
| <b>IRV</b>                   | <b>3.3L</b>                                                                                     | <b>2.2L</b>                                                                                         |
| <b>ERV</b>                   | <b>1.0 L</b>                                                                                    | <b>0.8 L</b>                                                                                        |
| <b>RV</b>                    | <b>1.5 L</b>                                                                                    | <b>1.3 L</b>                                                                                        |
| <b>TLC</b>                   | <b>6.5 L</b>                                                                                    | <b>4.8 L</b>                                                                                        |
| <b>VC</b>                    | <b>5.0 L</b>                                                                                    | <b>3.5 L</b>                                                                                        |
| <b>FRC</b>                   | <b>2.5 L</b>                                                                                    | <b>2.1 L</b>                                                                                        |
| <b>IC</b>                    | <b>4.0 L</b>                                                                                    | <b>2.7L</b>                                                                                         |



IRV = Inspiratory reserve volume      TV = Tidal volume  
ERV = Expiratory reserve volume      RV = Residual volume

| Volume (L)          |     |     |       |                              |
|---------------------|-----|-----|-------|------------------------------|
|                     |     | Men | Women |                              |
| Vital capacity      | IRV | 3.3 | 1.9   | Inspiratory capacity         |
|                     | TV  | 0.7 | 0.5   |                              |
|                     | ERV | 1.0 | 0.7   | Functional residual capacity |
|                     | RV  | 1.2 | 1.1   |                              |
| Total lung capacity |     | 6.0 | 4.2   |                              |

# Ventilation

## 1. Minute Ventilation

- The volume of air ventilated in a minute
- $MV = T_v \times f$  (tidal volume x Breathing frequency)
  - BF = 12 bpm (adults), 14-16 bpm (new born)

## 2. Alveolar Ventilation

- The rate at which new air reaches the gas exchange areas of the lungs (*respiratory bronchioles, alveoli, alveolar sacs, alveolar ducts*)
- $AV = (T_v - V_D) \times f$

# Ventilation

## 3. Dead space volume (VD)

- The volume of air which fills respiratory passages where gas exchange does not occur (nose, pharynx, trachea, bronchioles...)
- The air in the dead space is expired first during expiration
- Hence, disadvantageous for removing expiratory gases from the lungs

### A. Anatomic dead space

- Conduction air ways are fixed dead space
- It's volume  $\approx 150$  ml

# Ventilation

## B. Alveolar dead space

- Unperfused but ventilated alveoli (*increased by pulmonary embolism*)

## C. Physiologic dead space

- The sum of anatomic and alveolar dead space
- In a normal person, **physiologic VD  $\approx$  anatomic VD**

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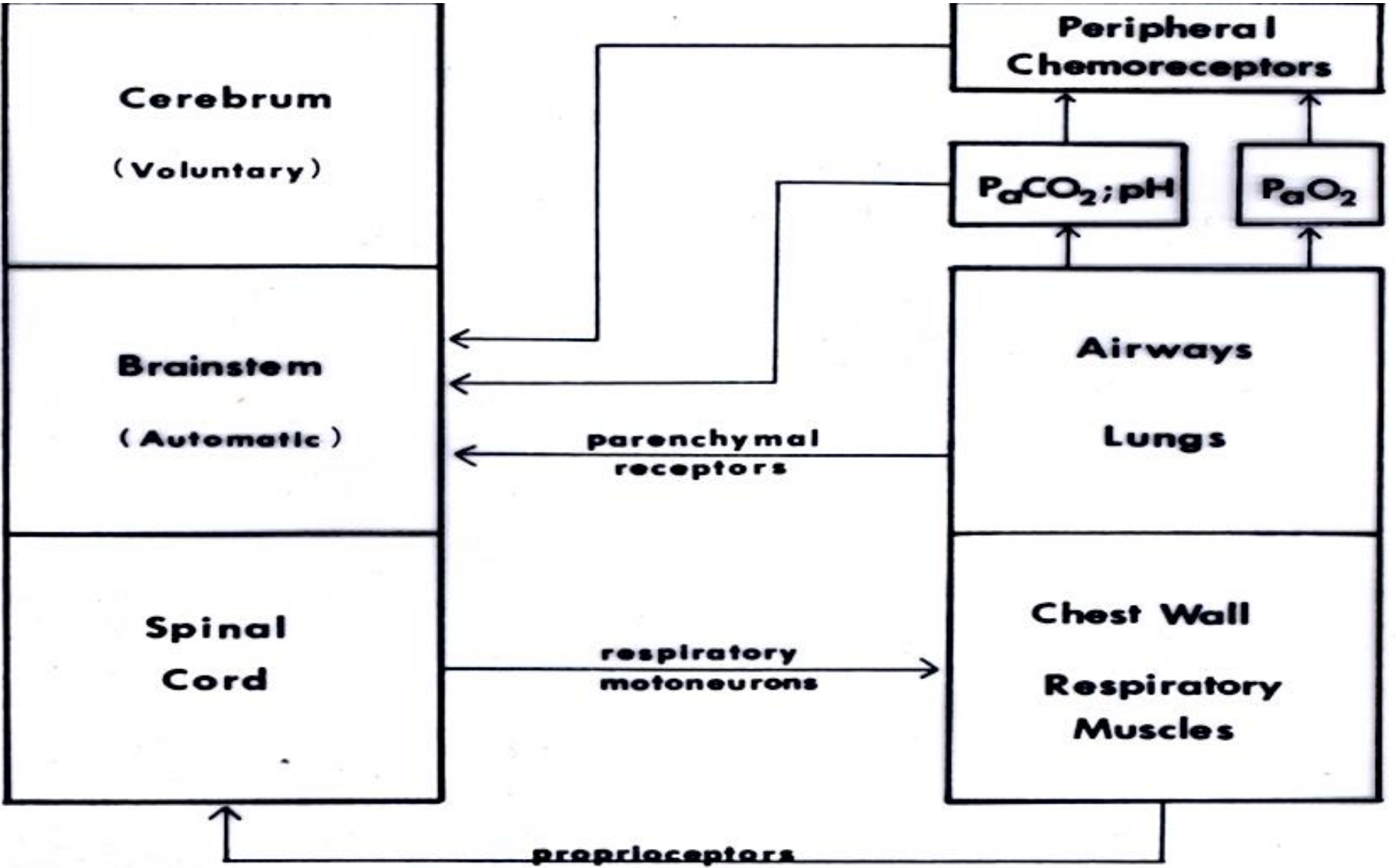
# **Regulation of Respiration**

# Respiratory Regulation

- Three basic components of Breathing control:
  1. Sensors (receptors) → peripheral and central
  2. Control centers → Neural and hormonal
  3. Effectors → Lungs, respiratory muscles
- Neural control is both **voluntary and involuntary**
- Neurons located in the Respiratory centers (pons and medulla) control the periodic nature of involuntary respiration → inspiration and expiration

# Respiratory Regulation

- **Cerebral Cortex:**
  - Responsible for coordinating breathing with voluntary activities
  - Can override the function of the brain stem within physiological limits
- **Automatic control mechanisms:**
  - Involves neurons of the brain stem
- **Reflex control mechanism:**
  - Requires chemoreceptors and mechanoreceptors situated outside the CNS



*Respiratory control system*

# Respiratory Receptors

## 1. Peripheral Chemoreceptors

- Found in **aortic + carotid bodies** in the neck region
- Detect changes in  $PO_2 + PCO_2 + [H^+]$
- Factors increasing activity of chemoreceptors:
  - $\downarrow PO_2$
  - $\downarrow pH$
  - $\downarrow$  Carotid BF
  - $\uparrow PCO_2$

# Respiratory Receptors

## 2. Central Chemoreceptors

- Found in the medullary region
- Factors increasing activity:
  - $\downarrow$ pH emanating from  $\uparrow$ PCO<sub>2</sub> in artery or CSF
  - $\uparrow$ CO<sub>2</sub>
- **NB:**
  - H<sup>+</sup> can not cross the blood brain barrier
  - Change of arterial pH due to CO<sub>2</sub> is reflected in the CSF pH change

# Respiratory Receptors

## 3. Ancillary Receptors

- Pulmonary Mechanoreceptors

### A. Stretch receptors:

- Found in smooth muscle of airways
- Respond to distention of lung
- **Inspiration** → ↑ lung volume → stimulation of stretch receptors from the chest and thoracic cage → ↑ frequency of vagal afferent impulses → inhibition of inspiratory centre → slowing of respiratory frequency due to increase in respiratory time → **Herring-Breuer inflation reflex**

# Respiratory Receptors

## B. Irritant receptors:

- **Rapidly adapting receptors** found between epithelial cells of airways, nose, nasopharynx, larynx, trachea
- Stimulated by noxious gases, cigarette smoke, inhaled dusts and cold air → brain stem → coughing, sneezing, bronchoconstriction

## C. J-receptors:

- Juxta-capillary in alveolar walls close to capillaries
- Respond to pulmonary congestion, edema, emboli or chemicals injected into pulmonary circulation

# Respiratory Receptors

## D. Ergo receptors

- Found in diaphragm, intercostal muscles
- Include tendon receptors and muscle spindles
- **Tendon receptors** → prevention of injury
- **Muscle spindles** sense stretching of muscles → control strength of contraction and adequate ventilation when there is ↑respiratory load

# Respiratory Receptors

## E. Thermoreceptors

### 1. Peripheral Thermoreceptors:

- Rise in body temperature → stimulation of peripheral thermoreceptors → signals through afferents → ↑ventilation

### 2. Central Thermoreceptors:

- Rise in body temperature → stimulation of hypothalamic thermoreceptors → ↑ventilation
- Pain often causes a period of apnea followed by hyperventilation

## Respiratory Centers

- Respiratory receptors in the brain stem receive afferent signals from respiratory receptors through sensory cranial nerves
- The Brain stem forms the breathing regulatory and coordinating centers consisting of a series of interconnected neurons:
  1. Medullary respiratory center (1°)
    - Dorsal Respiratory Group → DRG
    - Ventral Respiratory Group → VRG
  2. Pontine Respiratory Centers (2°)
    - Apneustic center + Pneumotaxic center

# Respiratory Centers

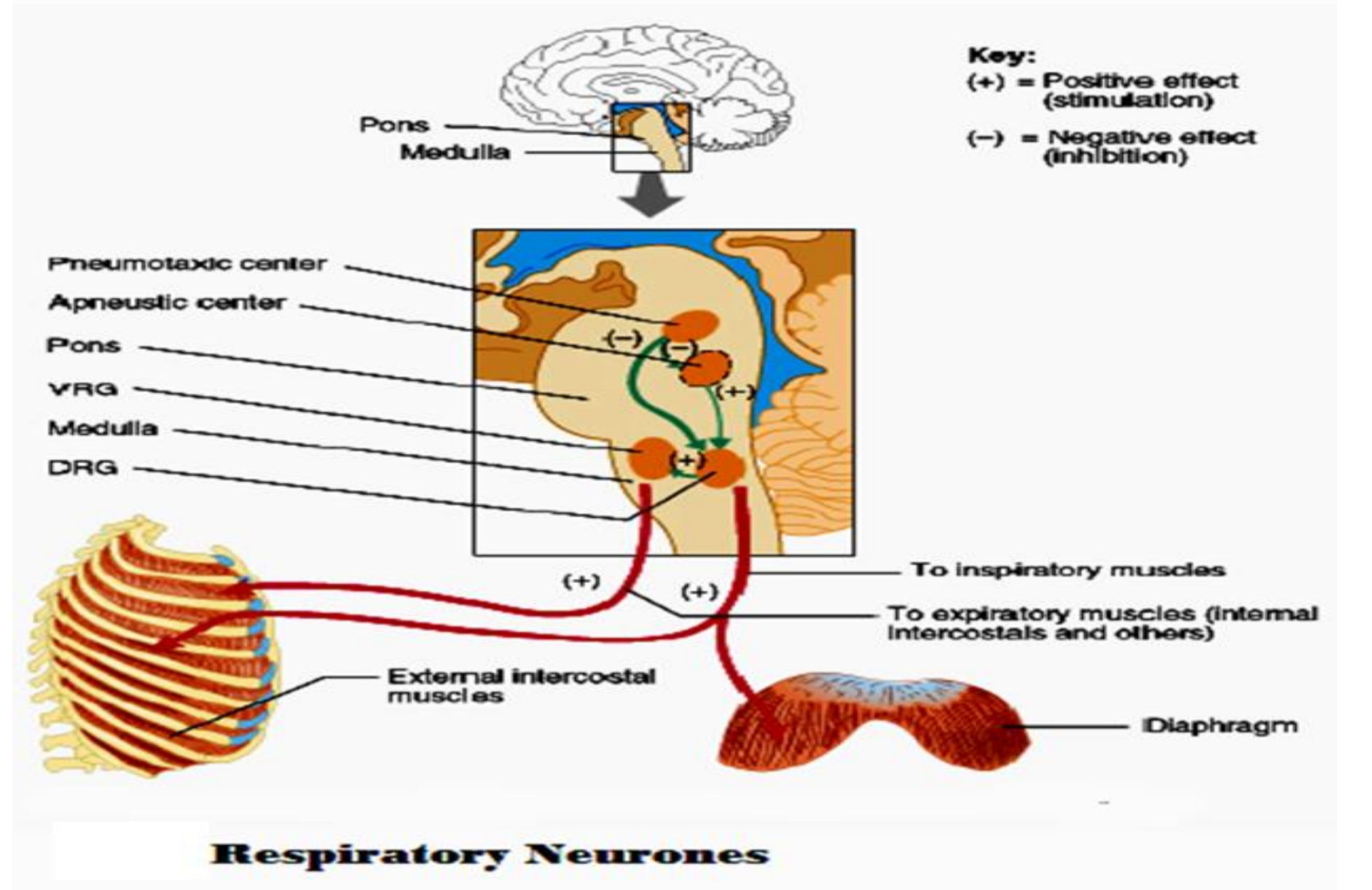
- Medullary respiratory group (VRG + DRG)
  - Constitute the **primary respiratory centers**
  - Initiate and maintain spontaneous respiratory patterns
  - Responsible for initiation and maintenance of spontaneous breathing patterns (*induce sustained inspiration + forced expiration*)
  - They are especially **sensitive during infancy** → the neurons can be destroyed if the infant is dropped or shaken violently → **Shaken Baby Syndrome**

# Respiratory Centers

- DRG
  - Primarily inspiratory and a major rhythm generator
  - Consists of Inspiratory Neurons located in & near the nucleus solitarius
- VRG
  - Mainly expiratory (*forced expiration*)
  - Consists of a long column of Expiratory Neurons extending through the nucleus ambiguus and the nucleus retroambiguus
  - These neurons remain almost totally inactive during normal quiet respiration

# Respiratory Centers

- Pontine Respiratory Center
  - **Apneustic**/gasping centre → lower and mid pons
  - **Pneumotaxic**/inhibitory centre → rostral part of pons
    - Transmits inhibitory signals to the inspiratory area → control the “switch-off” point of the inspiratory ramp → control the duration of the filling phase of the lung cycle
    - Primarily *limit inspiration*



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# **Respiratory Regulation of Acid-base Balance**

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## Acid-Base Balance

- Acid-base status of the blood is profoundly affected by the transport of CO<sub>2</sub>

- Respiration regulates PCO<sub>2</sub> and hence H<sup>+</sup>



- ↑Ventilation shifts the equilibrium to the left
- ↓Ventilation shifts equilibrium to the right
- Lungs excrete about 10,000mEq of acid compared with less than 100mEq of fixed acids excreted by the kidney
  - Thus, alteration of V<sub>A</sub> have a great influence on acid-base balance

# Acid-Base Disturbance

- **Respiratory Acidosis**
  - Blood  $[\text{H}_2\text{CO}_3] > 1.4\text{mEq/L}$
  - $\text{PaCO}_2$  is normally kept at or near 40 torr (normal range = 35-45 torr) by mechanisms that regulate breathing
- **Causes:**
  - Severe emphysema
  - Barbiturate poisoning
  - Breathing air containing high  $\text{CO}_2$
- **Effects:** muscle weakness, fatigue, drowsiness and coma

# Acid-Base Disturbance

- Respiratory Acidosis
  - Compensation:
    - Renal reabsorption of  $\text{HCO}_3^-$
    - Renal manufacturing of more  $\text{HCO}_3^-$
    - Use of respiratory stimulant

# Acid-Base Disturbance

- Respiratory Alkalosis
  - Blood  $[\text{H}_2\text{CO}_3] < 1.4\text{mEq/L}$
  - Causes:
    - Hyperventilation
    - Salicylate poisoning  $\rightarrow$  hyperventilation 2° to medullary stimulation
    - High fevers and head injury

# Acid-Base Disturbance

- Respiratory Alkalosis
  - Effects:
    - Muscle pain, twitching
    - Loss of appetite, headache, irritability, nausea and vomiting
  - Compensation:
    - ↑ Excretion of bicarbonate, administration of 5% CO<sub>2</sub>

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# **Drugs Acting on the Respiratory System**

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# Bronchodilators

## A. Anticholinergics

- Atropine methylnitrate (potent)
- Atropine sulphate

## B. Sympathomimetics

- Adrenaline – powerful bronchodilator
- Ephedrine - less potent
- Isoprenaline - powerful
- Orciprenaline (Alupent)
- Salbutamol (Ventolin)
- Terbutaline (Bricanyl)

## Bronchodilators

- Bronchodilators are used in treating bronchial asthma
- Corticosteroids are also given to patients with acute bronchial asthma and status asthmaticus
- **Bronchoconstrictors**
  - Histamine
  - Bradykinin
  - Prostaglandin F<sub>2a</sub>, F<sub>2b</sub>, etc.

## Respiratory Stimulants

- **Analeptics** (strengthening or restorative)
  - Nickethamide (coramine)
    - Stimulates all levels of CNS but does not produce respiratory stimulation in doses that cause little central excitation
  - Ethamivan (vandid)
  - Doxapram (dopram)

## Cough Suppressants

- Mainly opiates and opiate derivatives
  - Codeine 15mg
  - Dextromethorphan-10mg
  - Methadone-2.5mg
  - Noscapine-15mg
  - Pholcodeine 10mg
- **Mucolytic agents**
  - Bromhexine (bisolvon)
  - 5-carboxymethylcysteine (mucodyne)